

# Technical companion to

*Long runs and rare patterns of a random  
completely multiplicative function*

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Version 3

This research received substantial assistance from AI systems; see the [declaration at the end of the article](#).

## Abstract

This companion completes the proof package for the article, Version 3. Appendices A–E give the growing-degree Runge and bounded-height Pell calculations, prime cutoffs, scalar smoothing, marked and dictionary comparisons, conditional kernels, and the boundary and relative-bulk estimates. The common macroscopic square-product argument remains in the article.

Appendix F proves the full same-grid signed prefix comparison. It assembles the existing arithmetic inputs and an independently proved scalar tail, then gives the finite Palm–Stein inequality, the uniform reciprocal-pivot estimate and the exact one-word conditional coupling. Appendix G is a separate methodological complement: regular target configurations, signed Palm-void identities and an arithmetic obstruction to global absolute-cumulant smallness. Its vanishing Palm deficit is a consequence of Appendix F, never an input to its proof.

**Reading convention.** The two documents form one proof package. Appendix letters and bibliographic numbers are local; unqualified numbered statements refer to the article. All unlabelled logarithms are natural. The scope, explicit literature inputs and limitations of the Lean development, together with its seven Palomar registrations, are described in the article’s section “[Formalization and Palomar registrations](#)”.

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## External inputs and internal deductions

This register identifies the non-elementary inputs, their precise role, and the calculations proved in this package. References are local to this companion; statement links lead to the relevant document. Classical finite-dimensional linear algebra, conditioning and elementary inequalities are not separate imported hypotheses. The equation numbers (2.14)–(2.15) cited for Krokowski refer to arXiv version 3; the cited Lemma 4 of Cilleruelo–Garaev refers to their arXiv text.

| Input and source                                                                             | Form consumed                                                                          | Use and internal deduction                                                                                                                                           |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prime number theorem; standard prime sums [18]                                               | $\pi(t) \sim t/\log t$ ; elementary weighted consequences                              | Proposition B.1 proves both cutoff balances; Lemma F.3 and proposition F.5 prove the uniform reciprocal-pivot estimate.                                              |
| Finite-dependency Poisson approximation [1, 2]                                               | Scalar and process theorems; $b_3 = 0$ for an exact graph                              | <a href="#">Theorems 4.1, 5.1, and 5.8</a> ; the supremum convention for variation is half the sources' norm.                                                        |
| Scalar Stein factors [12, (2.14)–(2.15)]                                                     | Zeroth and first differences, of orders $\lambda^{-1/2}$ and $\lambda^{-1}$            | Lemma B.2 proves the exceptional-set decomposition; no Malliavin input is used.                                                                                      |
| Directional Poisson factor [4, p. 179, Lemmas 2–3]                                           | Weighted quadratic Hessian bound and $ \Delta_{ij}g  \leq 1$ , for the same solution   | Equation (C.3) and Theorem C.1; Hessian conjugation, independent Poisson filling and geometric mark summation are explicit.                                          |
| Palm–Stein method [4, 8]                                                                     | Immigration–death generator and conditional-law identity                               | Lemma F.2 proves the finite total-variation bound with no graph assumption; the exact arithmetic forcing is proved in proposition F.7.                               |
| Quadratic ideals and units [11]; divisors [14]                                               | Ideal factorization, unit orbits and $\tau(n) = \exp(O(\log n/\log \log n))$           | Lemma A.2 gives a self-contained proof of the classical uniform count; split-product localization is internal.                                                       |
| Uniform polynomial-height Pell count [9, arXiv text, Lemma 4]; [17, Lemma 2.10, Remark 2.12] | $\exp(O_K(\log M/\log \log M))$ solutions in polynomial-height boxes                   | The estimate itself has these antecedents; Appendix A.2 fixes the normalization and height dependence needed here.                                                   |
| Largest-prime reciprocal asymptotics [10, eq. (1.3)]                                         | $S_+(x) = x \exp\{-2V_x + o(\nu_x)\}$ at the precision used here                       | Appendix F.3 deduces the odd-pivot bound by $n = a^2v$ . The direct Rankin proof is retained; no separate analytic priority is claimed.                              |
| Laishram–Shorey [13, eq. (14), p. 331]                                                       | $\omega(\prod_{j < B}(n+j)) \geq (2-\varepsilon)\pi(B)$ , uniformly for $n > B$        | Appendix E.2 records the quantifiers; the private-coordinate and medium-prime rank arguments are internal.                                                           |
| Balasubramanian–Shorey [3], as recorded in [16, pp. 55–56]                                   | Square-product exclusion with $P^+(b) \leq k$ and the stated height/density conditions | Lemma E.3 gives the substitutions and contrapositive. No size bound on the smooth coefficient is added.                                                              |
| Berry's normal approximation bound [5]                                                       | Third-absolute-moment normal approximation                                             | Appendix D uses the explicit Poisson specialization; the marked comparison and joint limiting independence are proved here.                                          |
| Binary marker construction [7]                                                               | A classical cross-bifix-free family                                                    | <a href="#">Corollary 5.6</a> proves the particular binary construction and its size bound. This illustrative application is not an input to the arithmetic theorem. |

Runge’s split-product estimate, the Hamming-ball argument, the raw and capped two-window profiles, the simple-incidence rank lemma and the relative bulk normalization are proved in the article–companion pair. Evertse–Silverman is contextual only. The selected-pivot comparison and Palm complements are proved in Appendices F–G, with the latter using, not establishing, the microscopic comparison. The table concerns the mathematical inputs to the written arguments. In particular the directional estimate is obtained directly from Barbour’s two lemmas, while the arithmetic comparison and Poisson filling are proved here.

**Written inputs and formal coverage.** This table describes the written proof. The scope and limitations of the Lean development, including its seven explicit literature inputs, are recorded in the article’s section “[Formalization and Palomar registrations](#)”. In particular, the directional input supplies the entrywise bound and the weighted quadratic bound for the same Stein solution. An entrywise bound by one is not an unweighted Euclidean operator-norm bound. The derivation in (C.3) uses the weighted and entrywise estimates.

## How to use this companion

This companion and the article form a single proof package. The article gives the objects, main arguments and conclusions; the entries below identify the calculation behind each excursion and the point at which the main argument resumes. Unqualified numbers refer to the article; lettered numbers refer to this companion.

| Entry from the article                          | Calculation here                                                                      | Return to the article                                                                  |
|-------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <a href="#">Lemma 2.3</a>                       | Coefficients, integrality and the growing-degree remainder in Lemma A.1.              | Weighted defects, <a href="#">proposition 2.4</a> .                                    |
| <a href="#">Lemmas 3.11 and 3.12</a>            | Uniform ideal and unit counts, then localization to two square classes in A.2–A.3.    | Bounded components and the deep-core estimates.                                        |
| <a href="#">Proposition 4.2 and Theorem 4.3</a> | Weighted prime sums, degree bounds and scalar retention in B.1–B.3.                   | Scalar laws or the fields of Section 5.                                                |
| <a href="#">Theorems 5.1 and 5.3</a>            | Marginal cap, selection probabilities and collision bounds in C.1.                    | Uniform, typical and affine dictionaries.                                              |
| <a href="#">Theorem 5.10(ii)</a>                | The two Stein bounds, Poisson filling and the single tail event in C.3–C.4.           | The staircase; information-adapted labels in C.5 and <a href="#">proposition 6.2</a> . |
| <a href="#">Theorem 6.4</a>                     | Poisson atom estimates and the resolution cost in D.1–D.2.                            | Conditional laws in Section 6.                                                         |
| <a href="#">Theorem 7.1</a>                     | Prime incidences, quotient rank and the post-quadratic bound in E.1–E.5.              | The microscopic mass and the prefix law.                                               |
| <a href="#">Theorems 7.9 and 7.11</a>           | The normalized error table, finite-cylinder clock and affine conditioning in E.6–E.7. | The two-source comparison and its affine variant.                                      |

**The prefix coupling and its optional complement.** [Theorem 7.7](#) enters Appendix F. The signed input register Lemma F.1 points to the one-window and two-window estimates, local signed ranks, deep-start bound and independently proved scalar tail. The argument then proceeds through the finite Palm–Stein inequality, uniform Rankin estimate, reciprocal pivots and exact forcing. It returns to the full same-grid field and [Corollary 7.8](#). The restriction to

the dyadic field in [Corollary 7.7a](#) needs only the stability of the hard scales under  $M = 4N$  and projection of integer labels. No result of Appendix G is used along this route. Appendix G separately proves regular-cloud statements, finite signed Palm identities and the absolute-cumulant obstruction. Its vanishing Palm deficit is downstream of Appendix F. The empirical consequence [Corollary 7.8a](#) is proved in the article, immediately after [Corollary 7.8](#): it uses concentration of overlapping-window frequencies in the independent target and the summable microscopic error, with no additional arithmetic input. [Remark 7.8b](#) then gives a finite-support obstruction to the analogous local-vector comparison after fixing  $f$ .

**Support and target conventions.** For interior starts, the following probabilities hold on good supports conditionally on the exposed prime signs.

| Event                            | Values      | Independent equations | Target probability |
|----------------------------------|-------------|-----------------------|--------------------|
| Run of length at least $L$       | $L + 1$     | $L$                   | $2^{-L}$           |
| Absolute word of length $B$      | $B$         | $B$                   | $2^{-B}$           |
| Exact run, excess $e$            | $L + e + 2$ | $L + e + 1$           | $2^{-L-e-1}$       |
| Exact run, excess $e$ , sign $s$ | $L + e + 2$ | $L + e + 2$           | $2^{-L-e-2}$       |

Summing the last row over signs gives  $2^{-L-e-1}$ , and then summing over  $e \geq 0$  gives  $2^{-L}$ . With  $e \leq E$ , the maximal support has  $Q + 1$  values,  $Q = L + E + 1$ . This is distinct from the base support  $B = L + 1$  and from the exponential profile parameter  $Q_B = 2^B$ .

**Two different summations.** For the graph-based spatially labelled comparison of Section 5, marks are summed without weights before the pair estimate, leaving the relation profile at the base length. For an aggregated vector, the directional Stein weights are inserted first, and the maximal-length profile is required. Section C.3 keeps this distinction explicit before any asymptotic absorption. Its signed local factor can be four, not two. Full marks of base-contained starts are not right-censored run lengths.

In the selected-pivot comparison of Appendix F the signed low types are instead controlled on maximal supports and summed with their exact weights. Its dimension-free bound has no inverse minimum type probability. This is a third ledger, not a use of an aggregated Stein factor on an unrecorded spatial field.

## A Runge coefficients and uniform Diophantine inputs

This appendix supplies the growing-degree Runge calculation and the bounded-height Pell argument used in the article. The two types of uniformity are different: the Runge degree is allowed to grow, whereas a component's degree and its polynomial-height exponent are fixed before the ambient scale. No height-free integral-point estimate is needed.

### A.1 The split-product calculation

**Lemma A.1** (Coefficient-level Runge estimate). *Under the hypotheses of [Lemma 2.3](#), including  $U \in \mathbb{Z}$ , the bound  $U \leq (C_0 d R)^{2d}$  holds with an absolute constant  $C_0$ .*

*Proof.* The  $\gamma_\nu$  being distinct and  $d \geq 2$ , we have  $R \geq 1$  and  $G$  is not the square of a polynomial. Set

$$F(z) = \prod_{\nu=1}^d (1 + \gamma_\nu z)^{1/2} = \sum_{m \geq 0} c_m z^m,$$

where each square root is the analytic branch equal to 1 at 0, and

$$P(T) = \sum_{m=0}^k c_m T^{k-m}.$$

For  $m \leq k$ , the identity

$$\binom{1/2}{j} = (-1)^{j-1} \frac{\text{Cat}_{j-1}}{2^{2j-1}} \quad (j \geq 1)$$

implies  $2^{2j} \binom{1/2}{j} \in \mathbb{Z}$  and  $|\binom{1/2}{j}| \leq 1$ . More explicitly,

$$c_m = \sum_{j_1 + \dots + j_d = m} \prod_{\nu=1}^d \binom{1/2}{j_\nu} \gamma_\nu^{j_\nu}.$$

The number of weak compositions of  $m$  into  $d$  parts is  $\binom{m+d-1}{d-1} \leq 2^{m+d}$ . Since  $m \leq k = d/2$  and  $|\gamma_\nu| \leq R$ , we obtain simultaneously

$$2^d c_m \in \mathbb{Z}, \quad |c_m| \leq 2^{m+d} R^m \leq (8R)^d, \quad 0 \leq m \leq k. \quad (\text{A.1})$$

In particular  $2^d P \in \mathbb{Z}[T]$ , uniformly as  $d$  grows with the ambient scale.

If  $2R \leq U \leq 4R$ , the conclusion is immediate after enlarging  $C_0$ . Assume therefore  $U > 4R$ . On the circle  $|z| = (2R)^{-1}$ ,

$$|F(z)| \leq (3/2)^{d/2} =: M_d.$$

Cauchy's formula gives  $|c_m| \leq M_d (2R)^m$ . Since  $2R/U < 1/2$ ,

$$\begin{aligned} |\sqrt{G(U)} - P(U)| &= U^k \left| \sum_{m \geq k+1} c_m U^{-m} \right| \\ &\leq U^k M_d \sum_{m \geq k+1} (2R/U)^m \\ &\leq \frac{2M_d (2R)^{k+1}}{U} \leq \frac{(C_1 d R)^d}{U}. \end{aligned} \quad (\text{A.2})$$

If the difference is nonzero,  $\sqrt{G(U)}$  is an integer and, by (A.1),  $P(U) \in 2^{-d}\mathbb{Z}$ . It is therefore at least  $2^{-d}$  in absolute value. Inequality (A.2) then implies

$$U \leq 2^d (C_1 d R)^d \leq (C_0 d R)^{2d}.$$

Assume finally  $\sqrt{G(U)} = P(U)$ . The polynomial

$$Q(T) = 2^{2d} G(T) - (2^d P(T))^2$$

has integer coefficients. It is nonzero, since  $G$  is not a square. The expansion at infinity gives  $\sqrt{G(T)} - P(T) = O(T^{-1})$ , whence

$$\deg Q \leq k - 1.$$

The coefficient of  $T^r$  in  $G$  is a symmetric polynomial containing at most  $\binom{d}{r} \leq 2^d$  monomials, each of modulus at most  $R^d$  after the uniform upper bound  $R \geq 1$ . By (A.1), every coefficient of  $2^d P$  is at most  $(16R)^d$  in modulus. A coefficient of  $(2^d P)^2$  is the sum of at most  $d+1$  products

of two such coefficients. Consequently

$$H(2^{2d}G) \leq (8R)^{2d}, \quad H((2^dP)^2) \leq (d+1)(16R)^{2d},$$

and therefore, for an absolute constant  $C_2$ ,

$$H(Q) \leq (C_2dR)^{2d}.$$

This estimate is uniform in  $d$ ; it hides no step where  $d$  would be fixed. An integer zero of a nonzero integer polynomial has modulus at most  $1 + H(Q)$ , by the elementary Cauchy bound. After enlarging  $C_0$ , we again obtain (2.2).  $\square$

The integrality of  $U$  is used precisely in the separation by  $2^{-d}$ . Without it the statement is false: for  $G(T) = T(T+1)$  the real numbers  $U_m = (\sqrt{1+4m^2} - 1)/2$  tend to infinity and satisfy  $G(U_m) = m^2$ .

## A.2 Pell equations with varying coefficients

The uniform polynomial-height count below is a classical Pell estimate; see [9, arXiv text, Lemma 4] and [17, Lemma 2.10 and Remark 2.12]. For comparison with  $x^2 - Dy^2 = E$ , multiply  $Az^2 - Cw^2 = e$  by  $A$  and set  $x = Az$ ,  $D = AC$ ,  $E = Ae$ . The nonsquare and polynomial-height conditions are preserved. The proof below is retained to expose its uniformity, not as a claim that this counting result is new.

**Lemma A.2** (Uniform Pell count in a polynomial range). *Fix  $K_0 > 0$ . Let  $A, C$  be positive squarefree integers such that  $A/C$  is not a rational square, and let  $e \neq 0$ . Assume*

$$A, C, |e| \leq M^{K_0}.$$

*The number of integer solutions  $(z, w)$  of*

$$Az^2 - Cw^2 = e, \quad |z|, |w| \leq M^{K_0},$$

*is*

$$\exp\left(O_{K_0}\left(\frac{\log M}{\log \log M}\right)\right). \quad (\text{A.3})$$

*Proof.* Set  $g = (A, C)$ ,  $A = gA_0$ ,  $C = gC_0$ . Since  $A, C$  are squarefree,  $A_0, C_0$  are squarefree, coprime, and their prime supports are disjoint. If a solution exists,  $g \mid e$ ; write  $e = ge_0$ . After multiplying by  $A_0$  and substituting  $X = A_0z$ , the equation becomes

$$X^2 - Dw^2 = m_0, \quad D = A_0C_0, \quad m_0 = A_0e_0. \quad (\text{A.4})$$

The product  $D$  is squarefree. The hypothesis on  $A/C$  gives  $D > 1$ .

In the real quadratic field  $K_D = \mathbb{Q}(\sqrt{D})$ , using the standard ideal-and-unit description of quadratic orders [11], the element

$$\alpha = X + w\sqrt{D}$$

belongs to the ring of integers  $\mathcal{O}_{K_D}$ , since  $\sqrt{D}$  is an algebraic integer. Its norm is  $m_0$ . Consequently

$$(\alpha)(\bar{\alpha}) = (m_0),$$

and the integral ideal  $(\alpha)$  divides the ideal  $(m_0)$ . For each rational prime  $p^a \parallel m_0$ , the number of choices for the part above  $p$  is at most  $(a+1)^2$ : it equals  $(a+1)^2$  if  $p$  splits,  $a+1$  if it is inert,

and  $2a + 1 \leq (a + 1)^2$  if it ramifies, since then  $(p) = \mathfrak{p}^2$ . The number of ideal divisors of  $(m_0)$  is therefore at most

$$\prod_{p^a \parallel m_0} (a + 1)^2 = \tau(|m_0|)^2.$$

Only the principal ideals  $\mathfrak{a} = (\alpha)$  of absolute norm  $|m_0|$  that admit a generator of norm exactly  $m_0$  can come from a solution of (A.4). Fix such an ideal and a generator  $\alpha_0$  of norm  $m_0$ . Then

$$|\sigma_1(\alpha_0)\sigma_2(\alpha_0)| = |m_0|.$$

All relevant generators are among

$$\pm \alpha_0 \varepsilon_D^n, \quad n \in \mathbb{Z},$$

where  $\varepsilon_D > 1$  generates the free part of the unit group. If a unit of norm  $-1$  exists, the norm- $m_0$  condition selects only one parity class of  $n$ , which cannot increase the number of solutions.

The bounds on  $A, C, z, w$  give, for the two real embeddings  $\sigma_1, \sigma_2$ ,

$$|\sigma_j(\alpha)| \leq H := M^{O_{K_0}(1)}.$$

The inequalities

$$|\sigma_1(\alpha_0)|\varepsilon_D^n \leq H, \quad |\sigma_2(\alpha_0)|\varepsilon_D^{-n} \leq H$$

define an interval of integers  $n$  of length at most

$$\frac{2 \log H - \log |m_0|}{\log \varepsilon_D} + O(1) = O_{K_0}(\log M),$$

since every real quadratic unit  $\varepsilon > 1$  satisfies  $\varepsilon \geq (1 + \sqrt{5})/2$ . Indeed its conjugate is  $\pm \varepsilon^{-1}$  and its trace is an integer. For norm  $-1$ , this gives  $\varepsilon - \varepsilon^{-1} \geq 1$ ; for norm  $+1$ ,  $\varepsilon + \varepsilon^{-1} \geq 3$ . Both give the asserted lower bound. Thus the number of solutions is at most

$$O_{K_0}(\tau(|m_0|)^2 \log M).$$

Since  $|m_0| \leq M^{O_{K_0}(1)}$ , the standard explicit divisor bound [14] gives

$$\tau(|m_0|)^2 \log M = \exp\left(O_{K_0}\left(\frac{\log M}{\log \log M}\right)\right),$$

which proves (A.3); the sign of  $m_0$  does not affect the argument.  $\square$

The exponent  $K_0$  is fixed in every use. There is no appeal to a uniform lower bound for a regulator that deteriorates with the discriminant: the elementary absolute lower bound on a real quadratic fundamental unit already controls the number of generators in each ideal class.

### A.3 Square-class localization in bounded height

In Lemma 3.12, the decisive set is  $S = \{p : p \mid e \prod_{i < j} (h_i - h_j)\}$ . For a solution, a prime outside  $S$  can divide only one shifted factor; its valuation there is therefore even. We enumerate the squarefree parts of only the first two factors, not of all  $d$  factors. There are at most  $4^{|S|}$  pairs, and the two-factor equation has a nonzero right-hand side because  $h_1 \neq h_2$ .

Here is an explicit check on the order of parameters. Fix  $d, K$  first. For  $M \geq 2$ ,

$$\log |e\Delta| \leq K \log M + \binom{d}{2} (K \log M + \log 2) = O_{d,K}(\log M).$$

Thus  $4^{|S|} = \exp(O_{d,K}(\log M / \log \log M))$ . Independently, every retained squarefree coefficient is at most  $2M^K$ , not an unrestricted product of primes in  $S$ . The coefficients, right-hand side and square variables of its Pell equation are therefore all covered by the fixed exponent  $K + 1$  (increasing the ambient threshold if needed). The Pell constant does not depend on  $S$ , the offsets or the chosen pair. When the squarefree coefficients agree, divisor factorization treats the case that the nonsquare Pell lemma deliberately excludes.

For a component,  $d = |J| \leq K_0$  and the normalizing coefficient is the squarefree part of  $d_C \prod_{i \in I} (x + i)$ . The smooth coefficient  $d_C \leq \exp(O(B))$  and  $|I| \leq K_0$  put this integer in one fixed polynomial range. This supplies exactly the uniformity consumed by [Lemma 3.14](#), even when the two starts have different scales.

#### A.4 Why a single rational unit is not contracted

Two units force the reduced channel height to be at most  $L$ . One unit does not. For example, take  $N = 2^{100}$ ,  $L = 100$ ,  $a = 103$ ,  $b = 102$  and  $t = 12427947061061072563693168687$ , with  $x = bt$ ,  $y = at$ . Both starts lie in  $[N, 2N)$ , and  $103i = 102j$  has only  $(i, j) = (0, 0)$  in  $\{-1, \dots, 99\}^2$ . Yet  $103 > B = 101$ , and  $t \equiv 100 \pmod{103}$  gives  $v_{103}(x) = 0$ ,  $v_{103}(y) = 1$ . No vertex of the first window is divisible by 103, so the second occurrence is pinned rather than isolated with the first. This is why the article defines  $\sigma = 0$ ,  $D^\# = D$  and  $c^\# = c$  when  $m \leq 1$ . The aligned deep-core proof removes the at most two components meeting a single unit directly; it never assigns that unit a false private-kernel identity.

#### A.5 Uniformity across the master interval

All vertices are  $O(M)$  and at least  $M^\delta - 1$ . Squarefree  $B$ -smooth coefficients have size  $M^{O(1)}$ , while a fixed number of offsets and smooth choices contributes  $M^{o(1)}$ . On a larger-start terminal slice  $X \leq \max(x, y) < 2X$ ,  $B/\log X$  remains in a fixed compact interval depending only on  $(\delta, \beta_-, \beta_+)$ . Thus the Pell estimates can be applied at scale  $X$  there, even when the other start is smaller. The proof never identifies a macroscopic cross-block pair with an intra-block pair, and never applies a fixed-ratio theorem with a growing ratio.

## B Prime cutoffs and scalar smoothing

This appendix verifies the free cutoff before specializing it to either saddle. The distinction matters: the hard route deletes all exceptional starts, whereas the soft scalar route leaves them in the Stein equation.

### B.1 A uniform weighted prime sum

**Proposition B.1** (Free Rankin calculation and both saddles). *The bounds and expansions in [proposition 4.2](#) hold uniformly for  $w$  in a fixed multiplicative neighbourhood of  $\sqrt{H \log H}$ , where  $H = \log N$ .*

*Proof.* Every  $Y$ -defective integer has the unique form  $n = a^2 s$  with  $s$  squarefree and  $Y$ -smooth. Therefore, for  $0 < \zeta < 1/2$ ,

$$\frac{A(X, Y)}{X} \leq X^{-\zeta} \prod_{p \leq Y} (1 + p^{-1+\zeta}). \quad (\text{B.1})$$

Indeed sum  $\sqrt{X/s}$  over  $s \leq X$  and use  $s^{-1/2} \leq X^{1/2-\zeta} s^{-1+\zeta}$ . Put  $\nu = H/w$ , choose the larger solution  $u \geq 1$  of  $e^u/u = \nu$ , and set  $\zeta = u/w$ . Uniformly in the stated range,

$$u \sim \frac{1}{2} \log H, \quad \nu \asymp \sqrt{H/\log H}, \quad \zeta \asymp \sqrt{\log H/H}, \quad \log H = o(\nu).$$

For  $X/N$  in a fixed compact subset of  $(0, \infty)$ ,  $\zeta \log X = u\nu + o(\nu)$ .

We require the prime number theorem inside the weighted Euler product, not merely an upper bound with an unspecified constant. Let  $R(t) = \pi(t) - \text{Li}(t)$ . Given  $\epsilon > 0$ , choose a fixed lower split point beyond which  $|R(t)| \leq \epsilon t / \log t$ . Stieltjes integration by parts gives, uniformly in  $w$ ,

$$\begin{aligned} \sum_{p \leq e^w} p^{-1+\zeta} &= \int_2^{e^w} \frac{t^{-1+\zeta}}{\log t} dt + O_\epsilon(1) + O\left(\epsilon \frac{e^u}{w} + \epsilon \int_2^{e^w} \frac{t^{-1+\zeta}}{\log t} dt\right) \\ &= \text{Ei}(u) + o(\nu). \end{aligned}$$

For the second equality first let  $N \rightarrow \infty$  and then  $\epsilon \downarrow 0$ . The substitution  $z = \zeta \log t$  costs only  $O(\log(1/\zeta)) = o(\nu)$  at the lower endpoint, and  $\text{Ei}(u) \asymp e^u/u = \nu$ . Also  $\sum_p p^{-2+2\zeta} = O(1)$ , so logarithms of the Euler factors give

$$\sum_{p \leq e^w} \log(1 + p^{-1+\zeta}) = \text{Ei}(u) + o(\nu). \quad (\text{B.2})$$

Floors in  $Y = \lfloor e^w \rfloor$  alter these estimates by a negligible quantity. Equations (B.1)–(B.2) yield

$$\log(A(X, Y)/X) \leq -\nu u + \text{Ei}(u) + o(\nu) = -D(\nu) + o(\nu).$$

A bad integer contaminates at most  $B$  windows;  $\log B = o(\nu)$  absorbs this factor. The graph-edge estimate in the article gives  $E_Y/N^2 \leq e^{-w+o(\nu)} + N^{-1+o(1)}$ . A window uses at most  $CBH/w$  large primes, and each prime meets at most  $CB(N/Y + 1)$  starts. Hence

$$\Delta_Y \ll B^2(H/w)(N/e^w + 1), \quad \Delta_Y/N \leq e^{-w+o(\nu)} + N^{-1+o(1)}. \quad (\text{B.3})$$

This maximum-degree bound applies to the graph on all starts, including exceptional ones.

For a unified expansion write  $V_a = aD(H/V_a)$  for  $a = 1$  or  $2$ ,  $\nu_a = H/V_a$ , and  $u_a = u(\nu_a)$ . Since  $D' = u > 0$ , the defining function decreases strictly in  $V_a$ . At zero its sign is positive; at  $H/e$  its sign is negative for sufficiently large  $H$ . Thus the root is unique. The parametric identities are

$$\nu_a = e^{u_a}/u_a, \quad V_a = a(e^{u_a} - \text{Ei}(u_a)).$$

Using  $\text{Ei}(u) = e^u u^{-1}(1 + u^{-1} + O(u^{-2}))$  gives

$$\log H = 2u_a - \log u_a + \log a + O(u_a^{-1}), \quad u_a = \frac{1}{2}(\log H + \log \log H - \log(2a)) + o(1),$$

and  $V_a^2/H = a(u_a - 1 + O(u_a^{-1}))$ . Substitution gives (4.6) and (4.7).  $\square$

The second-order constants belong to this PNT main-term optimization. They are not matching lower bounds for the probability distance.

## B.2 A finite soft-exception Stein lemma

**Lemma B.2** (Retention of exceptional indicators). *Let a finite indicator family  $(I_\alpha)_{\alpha \in \mathcal{I}}$  have an exact dependency graph conditionally on  $\mathcal{F}$ . Split  $\mathcal{I} = \mathcal{G} \sqcup \mathcal{X}$  deterministically, and assume  $\mathbb{E}(I_\alpha | \mathcal{F}) = p$  for every  $\alpha \in \mathcal{G}$ . Put  $W = \sum I_\alpha$ ,  $\lambda = |\mathcal{I}|p$ ,  $\lambda_{\mathcal{X}} = |\mathcal{X}|p$  and  $W_{\mathcal{X}} = \sum_{\alpha \in \mathcal{X}} I_\alpha$ . Let  $p_\beta(\omega) = \mathbb{E}(I_\beta | \mathcal{F})(\omega)$ . The mean conditional Poisson distance is at most a constant times*

$$(1 \wedge \lambda^{-1}) \left[ |\mathcal{G}|p^2 + p \sum_{\alpha \in \mathcal{G}} \sum_{\beta \in B_\alpha \setminus \{\alpha\}} \mathbb{E} I_\beta + \sum_{\alpha \in \mathcal{G}} \sum_{\beta \in B_\alpha \setminus \{\alpha\}} \mathbb{E}(I_\alpha I_\beta) \right] + (1 \wedge \lambda^{-1/2})(\lambda_{\mathcal{X}} + \mathbb{E} W_{\mathcal{X}}). \quad (\text{B.4})$$

The two minimum factors are interpreted as one if  $\lambda = 0$ . No marginal assumption is made for exceptional neighbours.

*Proof.* If  $\lambda = 0$ , all good indicators vanish almost surely and  $d_{\text{TV}}(\mathcal{L}(W | \mathcal{F}), \delta_0) \leq \mathbb{E}(W_{\mathcal{X}} | \mathcal{F})$ ; averaging proves the claim. For  $\lambda > 0$  and a target test set  $A \subset \mathbb{N}_0$ , let  $f = f_{\lambda, A}$  solve  $\lambda f(k+1) - kf(k) = \mathbf{1}_A(k) - \text{Pois}(\lambda)(A)$ . The standard factors are

$$\|f\|_\infty \leq 1 \wedge \sqrt{2/(e\lambda)}, \quad \|\Delta f\|_\infty \leq (1 - e^{-\lambda})/\lambda \leq 1 \wedge \lambda^{-1}$$

[12, (2.15)]; see also [15, (2.4)]. For  $\alpha \in \mathcal{G}$  set  $U_\alpha = \sum_{\beta \in B_\alpha \setminus \{\alpha\}} I_\beta$  and  $V_\alpha = W - I_\alpha - U_\alpha$ . On each fibre  $I_\alpha$  is independent of  $V_\alpha$ . Add and subtract  $p\mathbb{E}[f(V_\alpha + 1) | \mathcal{F}]$ . Telescoping gives

$$|p\mathbb{E}[f(W+1) | \mathcal{F}] - \mathbb{E}[I_\alpha f(W) | \mathcal{F}]| \leq \|\Delta f\|_\infty \left[ p^2 + p \sum_{\beta \in B_\alpha \setminus \{\alpha\}} p_\beta + \sum_{\beta \in B_\alpha \setminus \{\alpha\}} \mathbb{E}(I_\alpha I_\beta | \mathcal{F}) \right].$$

The remaining term is bounded directly by

$$|\mathbb{E}[\lambda_{\mathcal{X}} f(W+1) - W_{\mathcal{X}} f(W) | \mathcal{F}]| \leq \|f\|_\infty (\lambda_{\mathcal{X}} + \mathbb{E}[W_{\mathcal{X}} | \mathcal{F}]).$$

The bound is uniform in the test set, so taking the supremum on each fibre and then averaging proves (B.4).  $\square$

## B.3 Applying the lemma at a movable cutoff

Take  $\mathcal{X} = D_Y$ ,  $p = 2^{-L}$  and  $\lambda = Np > 0$ . Set  $\ell = \log^+ \lambda$  and  $K = \max(1, \lambda)$ . The elementary inequalities

$$(1 \wedge \lambda^{-1})\lambda^2 \leq K, \quad (1 \wedge \lambda^{-1})(\lambda^2 + \lambda) \leq 2K, \quad (1 \wedge \lambda^{-1/2})\lambda \leq \sqrt{K}$$

hold on both sides of intensity one. For neighbour marginals use the maximum degree of the graph on all sites:

$$p \sum_{x \in \mathcal{G}} \sum_{y \in B_x \setminus \{x\}} \mathbb{E} J_{y,L} \leq p \Delta_Y \mathbb{E} Z_{N,L} \leq p \Delta_Y (\lambda + p \mathcal{M}_B).$$

The first-difference factor makes this at most  $K(\Delta_Y/N)(1 + N^{-1/2+o(1)})$ . The true marginals of exceptional neighbours have not been replaced by  $p$ . For the joint sum, overlapping starts contribute zero and the local bound treats touching pairs; the separated pairs give

$$\sum_{x \in \mathcal{G}} \sum_{y \in B_x \setminus \{x\}} \mathbb{P}(J_{x,L} J_{y,L} = 1) \ll p^2(N^{1+o(1)} + N\Delta_Y + R_2(N, L)).$$

Since  $R_2/N^2 \ll N^{-1/3+\varepsilon/4}(1 + \lambda^{-1})$ , its contribution, after the Stein factor, is at most  $KN^{-1/3+\varepsilon/2}$  after enlarging the threshold. The diagonal and local polynomial costs are smaller. Finally the exceptional zeroth-order term is at most

$$C\sqrt{K} \left( \frac{\mathcal{M}_B}{N} + \frac{|D_Y|}{N} \right) \leq Ce^{\ell/2 - D(H/w) + o(H/w)} + KN^{-1/3+\varepsilon/2}.$$

Together with (B.3), these estimates give (4.10) for all intensities.

At  $w = \widehat{V}_N$ , let  $\nu = H/w$  and choose a common  $\eta_N \geq 0$ ,  $\eta_N \rightarrow 0$ , dominating the displayed remainder terms. Put

$$q = e^{-(w-\ell)/2 + \eta_N \nu}, \quad r = e^{\ell - w + \eta_N \nu}, \quad P = N^{-1/3+\varepsilon/2}.$$

Up to an absolute constant, the error is at most  $q + r + KP$ . If  $q + N^{-1/3+\varepsilon} \geq 1$ , the trivial total-variation bound suffices. Otherwise  $q < 1$  implies  $\ell \leq w$  and  $r \leq q$ . Since  $w = o(\log N)$ , eventually  $K \leq e^w \leq N^{\varepsilon/2}$ , so  $KP \leq N^{-1/3+\varepsilon}$ . This proves the soft minimum bound, including  $\lambda < 1$ , at the same cutoff and on the full  $\mathcal{F}_Y$ . All thresholds depend only on the fixed band and the error margin, not on its length parameter. The numerical constants in a coarser ledger do not affect the claimed asymptotic bound.

All terms admit their conditional versions: integrate the fibrewise bound over  $C \in \mathcal{F}_Y$  and divide by  $\mathbb{P}(C)$ . A single factor  $e^I$  multiplies the ledger, where  $I = -\log \mathbb{P}(C)$ ; the leading soft exponent consequently has sufficient budget  $2I + \log^+ \lambda < \widehat{V}$ . This argument applies only to events measurable in the chosen prime field. It does not turn arbitrary post-selection on a run statistic into an arithmetic conditioning event.

## C Marked fields and spatial aggregation

The distinction between a labelled field and its aggregate is already visible in a finite family. This appendix gives the finite inequalities first; all passages to countably many marks are then controlled by a single tail event.

### C.1 Uniform and averaged dictionary comparisons

The marginal-ceiling calculation in the proof of Theorem 5.1 gives the finite inequality

$$\mathbb{E}\Delta_{\mathcal{W}} \ll a(\mathcal{M}_B^{\text{val}} + |D_Y^{\text{val}}|) + a^2(NB + E_Y + R_2^{\text{val}, [1/a]}(N, B)) + \Lambda\Omega(\mathcal{W}),$$

where  $a = m2^{-B}$  and  $\Lambda = Na$ . The comparison retains the full site–word index set; only the error terms have been summed over words. We verify here the substitutions and the uniform absorption of secondary terms. With  $T = N/\Lambda$  and  $Q_B = Nm/\Lambda \geq T$ , the three capped-profile terms become, respectively,

$$N^{-1/3+\varepsilon} \Lambda^{11/6} m^{1/6}, \quad N^{-1/3+\varepsilon} \Lambda^{4/3} m^{2/3}, \quad N^{-1/3+\varepsilon} \Lambda.$$

These substitutions are exact because  $a = \Lambda/N$  and  $Q_B = Nm/\Lambda$ . The restrictions  $1 \leq m \leq 2^B$  also give  $0 < a \leq 1$ , so the arithmetic cap  $T = 1/a$  is always at least one. The secondary edge contribution is absorbed in every admissible intensity regime, not only at criticality:  $m \geq \Lambda/N$  implies

$$\frac{\Lambda^2}{N} \leq N^{-1/3} \Lambda^{4/3} m^{2/3}.$$

Factors  $N^{o(1)}$  in these smaller terms are absorbed by fixing a smaller arithmetic error exponent first. At bounded positive  $\Lambda$  and  $m \leq N^{1/2-\eta}$ , the powers of  $N$  in the three terms, with  $\varepsilon = \eta/6$ , are at most

$$-\frac{1}{4}, \quad -\frac{\eta}{2}, \quad -\frac{1}{3} + \frac{\eta}{6} \leq -\frac{\eta}{2} \quad (0 < \eta < 1/2).$$

Thus all three are  $O(N^{-\eta/2})$ , uniformly in the dictionary. No independent random marks are added to the multiplicative sequence, and no word identity is lost when the error terms are summed.

### Averaging the finite dictionary comparison

We verify (5.14) before using any asymptotic profile. Fix  $N \geq 2$ ,  $B \geq 1$ ,  $Y > 2B$ , a deterministic start mask  $A \subset I_N$ , and a random size- $m$  dictionary,  $1 \leq m \leq 2^B$ , independent of the prime signs. Only the inclusion probabilities (5.11) are needed; both ensembles in the article have them. For each realization of the dictionary, apply the process inequality on the finite coordinate set  $A \times \mathcal{W}$ .

**Deletion and the exact graph.** Let  $D = D_Y^{\text{val}} \cap A$ . Removing real indicators has mean cost, after averaging the dictionary and the prime environment,

$$\sum_{x \in D} \mathbb{E}_f \mathbb{P}_{\mathcal{W}}(\mathbf{f}_x \in \mathcal{W}) = a|D|.$$

The Poisson target has total deleted mean  $a|D|$ . This uses no uniformity of a bad window's law. At a retained site every coordinate has conditional mean  $p_B = 2^{-B}$ . Its neighbours include all words at its own site, at each overlapping support and at each support sharing an odd prime coordinate above  $Y$ . Families with no edge depend on disjoint remaining prime signs, so their conditional independence is exact. The site graph and its good set do not depend on  $\mathcal{W}$ . The first Chen–Stein sum, including each diagonal coordinate, is bounded by  $Ca^2(|A|B + E_Y(A))$ .

**The second sum.** At the same site its off-diagonal terms vanish by mutual exclusivity; a coordinate's own diagonal belongs to the first sum only. For distinct sites the sum over word labels is exactly  $\mathbb{P}_f(\mathbf{f}_x, \mathbf{f}_y \in \mathcal{W} \mid \mathcal{F}_Y)$ . After both averages it is  $b + (a - b)\mathbb{P}_f(\mathbf{f}_x = \mathbf{f}_y)$ , as in (5.12). At a good local pair with displacement  $d < B$ , write the union values as  $z_0, \dots, z_{B+d-1}$ . The equations  $z_j = z_{j+d}$ ,  $0 \leq j < B$ , form the  $d$  paths with residue classes modulo  $d$ . There are  $B + d$  vertices and  $B$  edges. A prime above  $Y > 2B$  can occur at only one union vertex, and each vertex has such an odd coordinate, so the conditional vector is uniform. Its collision probability is exactly  $2^{-B}$  on every fibre.

For disjoint supports, let  $v(n)$  denote the complete binary valuation vector at the finite set of primes dividing the union. The equality matrix has rows  $v(x - 1 + j) + v(y - 1 + j)$ . Its coefficient kernel maps injectively by

$$t \longmapsto (t, t), \quad \sum_j t_j (v(x - 1 + j) + v(y - 1 + j)) = 0,$$

into the coefficient kernel of the stacked value matrix. The equality right-hand side is zero, so its probability is exactly  $2^{-B+\kappa}$ , regardless of whether either value window is individually uniform. This proves (5.13) with the full, not the large-prime-truncated, nullity. It is used only after averaging the prime fibres. Consequently a separated edge costs at most

$$a^2 + ap_B + ap_B(2^{\rho_B^{\text{val}}} - 1) \leq 2a^2 + ap_B(2^{\rho_B^{\text{val}}} - 1).$$

The local sum is  $O(a^2|A|B)$  and the separated sum is  $O(a^2E_Y(A) + ap_BR_2^{\text{val}}(A, B))$ . All site pairs here are ordered; the two orientations are included. Adding the two deletion costs proves (5.14) with an absolute implied constant. This finite inequality also covers  $m = 1$ ,  $B = 1$  and the empty mask.

**Normalization and the order of averaging.** With  $a = \Lambda/N$  and  $p_B = \Lambda/(Nm)$ , the two terms of the coarse profile give, separately,

$$ap_B N^{5/3} = \frac{\Lambda^2}{m} N^{-1/3}, \quad ap_B N^{2/3} 2^B = \Lambda N^{-1/3}.$$

The factor  $N^\varepsilon$  is inserted only after these exact identities. For an independent-letter word field, distinct disjoint supports need no edge, and every local collision again has probability  $p_B$ . Its averaged Poisson error is therefore  $O(a^2|A|B)$ , without a restriction on  $m$ . This is a fresh finite calculation, not an application of the uniform dictionary theorem outside its range.

The supremum over target tests is taken for each fixed dictionary and prime fibre *before* averaging their distances. Finite sums over the auxiliary dictionary law then give the claimed mean bound. The class of good dictionaries supplied by Markov controls all coordinate restrictions simultaneously by contraction. No mask may use the observed prime signs. A dictionary-dependent deterministic restriction is harmless once that dictionary has been selected. No coupling over different scales, or almost-sure choice of a good dictionary at every scale, is inferred.

## C.2 Local pairs and the role of the maximal support

From here on,  $p = 2^{-L}$ ,  $q_e = 2^{-e-1}$ ,  $p_e = pq_e$ ,  $\lambda = Np$  and  $0 \leq e \leq E$ . The largest row length is  $Q = L + E + 1$ ; its support has  $Q + 1$  vertices. An unsigned mark of excess  $e$  imposes  $L + e + 1$  relative equations; prescribing its sign fixes all  $L + e + 2$  values on the same support.

A good maximal support contains no  $Y$ -defective vertex and has diameter  $Q < Y$ . Every exact-mark system on that support consequently has full rank above  $Y$ , with conditional marginal  $p_e$ . If two actual supports intersect, their union has diameter at most  $2Q < Y$ ; every vertex of the union then has a prime coordinate private to that union. For two compatible exact runs the only possible intersecting cases are a common terminal/border vertex or a common terminal edge. Equivalently, for  $x < y$ , they satisfy  $y - x = L + e + 1$  or  $y - x = L + e$ . The union of the two constraint trees has cycle rank at most one, so its probability is at most  $2p_e p_f$ .

The actual supports can also be disjoint while the *maximal* supports intersect. In that case all their private coordinates are still distinct, and the probability is  $p_e p_f$ . Thus the bound

$$\mathbb{P}(K_{x,e} K_{y,f} = 1 \mid \mathcal{F}_Y) \leq 2p_e p_f \quad (0 < |x - y| \leq Q) \quad (\text{C.1})$$

holds for every good pair; it does not assert that only two separations are possible among pairs of intersecting maximal supports. At the same site, different exact marks are mutually exclusive.

For  $|x - y| > Q$ , zero-extension of the coefficient vectors embeds the mixed relation space into the two-window space at row length  $Q$ . Therefore

$$\mathbb{P}(K_{x,e}K_{y,f} = 1) \leq p_e p_f 2^{\rho_Q(x,y)}. \quad (\text{C.2})$$

This is an averaged, not necessarily fibrewise, estimate: the small-prime phase in the affine formula can vary with the environment.

For signed marks, put  $p_{e,s} = p_e/2$  and let  $a = L + e$ ,  $b = L + f$ . On a good local union the raw value vector is uniform. For  $x < y$ ,  $d = y - x$ , the exact joint probabilities are

$$\mathbb{P}(K_{x,e}^s K_{y,f}^t = 1 \mid \mathcal{F}_Y) = \begin{cases} 0, & d < a, \\ 4p_{e,s}p_{f,t}\mathbf{1}_{\{t=-s\}}, & d = a, \\ 2p_{e,s}p_{f,t}\mathbf{1}_{\{t=s\}}, & d = a + 1, \\ p_{e,s}p_{f,t}, & a + 1 < d \leq Q. \end{cases}$$

The final case concerns disjoint actual supports inside intersecting maximal supports. The factor four, rather than two, is needed when signs are prescribed. After summing signs one recovers (C.1). For separated signed systems, zero-extension instead gives the bound  $p_{e,s}p_{f,t}2^{\rho_{Q+1}^{\text{val}}(x,y)}$ . [Proposition 3.24](#) controls its weighted sum.

### C.3 The directional multivariate Stein factor

Let  $\pi = \bigotimes_{i=1}^d \text{Pois}(\lambda_i)$ , with  $d \geq 2$ ,  $\lambda_i > 0$ ,  $\lambda' = \sum_i \lambda_i$  and  $\mu_i = \lambda_i/\lambda'$ . For a test set  $A \subseteq \mathbb{N}_0^d$ , use Barbour's immigration–death solution, with its sign chosen so that

$$\mathcal{A}g(z) = \mathbf{1}_A(z) - \pi(A), \quad \mathcal{A}g(z) = \sum_i \{\lambda_i \Delta_i g(z) + z_i [g(z - \mathbf{e}_i) - g(z)]\}.$$

At  $z_i = 0$  the corresponding death term is zero. Write  $H(z) = (\Delta_{ij}g(z))_{i,j}$  and  $c(\lambda') = (1 + 2\log^+(2\lambda'))/2$ . Barbour's Lemmas 2 and 3 [4, p. 179] give, for this same solution,

$$|\Delta_{ij}g(z)| \leq 1, \quad |\alpha^T H(z) \alpha| \leq c(\lambda') \sum_i \frac{\alpha_i^2}{\lambda_i}.$$

The weighted bound, stated there for integer directions, extends to rational directions by homogeneity and to real directions by continuity. The literature input consists of the solution and these analytic bounds, not an arithmetic or aggregated total-variation estimate.

For clarity, the constant entrywise bound also follows directly from the solution's semigroup representation. Couple the processes started at  $z$ ,  $z + \mathbf{e}_i$ ,  $z + \mathbf{e}_j$  and  $z + \mathbf{e}_i + \mathbf{e}_j$  by adding two independent particles of rate-one exponential lifetimes. The second difference vanishes once either added particle dies; before that time the second difference of an indicator is at most two in absolute value. Its time integral is thus bounded by  $\int_0^\infty 2e^{-2t} dt = 1$ . The argument includes  $i = j$ .

To obtain the weighted entrywise bound, conjugate the symmetric matrix  $H(z)$  by  $D = \text{diag}(\sqrt{\lambda_i})$ . The quadratic inequality gives  $\|DH(z)D\|_{2 \rightarrow 2} \leq c(\lambda')$ . Bounding each entry by this norm, and using the constant bound above, yields

$$|\Delta_{ij}g(z)| \leq \min \left\{ 1, \frac{1 + 2\log^+(2\lambda')}{2\lambda' \sqrt{\mu_i \mu_j}} \right\} \leq \min \left\{ 1, \frac{C(1 + \log^+(2\lambda'))}{\lambda' \sqrt{\mu_i \mu_j}} \right\}. \quad (\text{C.3})$$

The logarithmic term in multivariate Stein factors is discussed in [15]; the two bounds used above are taken directly from Barbour. For the truncated geometric marks,  $S_E = \sum_{e=0}^E q_e \in [1/2, 1]$ ,  $\lambda' = \lambda S_E$  and  $\mu_e = q_e/S_E$ . The second expression is thus bounded by  $C(1 + \log^+(2\lambda))/(\lambda\sqrt{q_e q_f})$ . The summability that removes dependence on the number of marks is

$$\sum_{e \geq 0} \sqrt{q_e} = 1 + \sqrt{2}. \quad (\text{C.4})$$

**Theorem C.1** (Finite aggregated comparison). *Let  $X_e = \sum_{x \in I_N} K_{x,e}$ , and let  $D_Y(Q)$  be the bad maximal supports. Put*

$$s(\lambda) = 1 \wedge \frac{C(1 + \log^+(2\lambda))}{\lambda}.$$

*Uniformly while  $L + 1$  and  $Q + 1$  remain in a fixed positive logarithmic band and  $2Q < Y$ , one has*

$$\begin{aligned} \mathbb{E} d_{\text{TV}} \left( \mathcal{L}((X_e)_{0 \leq e \leq E} \mid \mathcal{F}_Y), \bigotimes_{e=0}^E \text{Pois}(\lambda q_e) \right) \\ \ll p\mathcal{M}_B + p|D_Y(Q)| + s(\lambda)p^2(NQ + E_Y(Q) + R_2(N, Q)). \end{aligned} \quad (\text{C.5})$$

Here  $\mathcal{M}_B$  is the one-window defect mass at the base length, and  $E_Y(Q)$  is the ordered large-prime edge count for maximal supports.

*Proof.* Write  $D = D_Y(Q)$  and retain the good-site counts  $G_e = \sum_{x \notin D} K_{x,e}$ . Independently of the arithmetic variables, introduce independent  $Z_e \sim \text{Pois}(\zeta_e)$  with  $\zeta_e = |D|pq_e$ , and put  $W = G + Z$ . Removing the actual bad-site marks and adding this Poisson filling changes the law by at most

$$\sum_{x \in D} \mathbb{P}(J_{x,L} = 1) + \sum_{e=0}^E \zeta_e \ll p\mathcal{M}_B + p|D|.$$

This is an averaged conditional coupling bound. It has no factor  $E$ , since the actual marks are disjoint refinements of  $J_{x,L}$  and  $\sum_{e \leq E} q_e \leq 1$ .

Use the Stein solution for the *full* target means  $\lambda q_e$ . For the coordinate vector  $\mathbf{e}_e$  and  $\Delta_e g(w) = g(w + \mathbf{e}_e) - g(w)$ , Poisson summation gives, conditionally on the good indicators and  $\mathcal{F}_Y$ ,

$$\mathbb{E}[Z_e \Delta_e g(W - \mathbf{e}_e) \mid G, \mathcal{F}_Y] = \zeta_e \mathbb{E}[\Delta_e g(W) \mid G, \mathcal{F}_Y].$$

Indeed  $z\mathbb{P}(Z_e = z) = \zeta_e \mathbb{P}(Z_e = z - 1)$  for  $z \geq 1$ ; the term with  $Z_e = 0$  is zero. Thus the filling terms cancel the  $\zeta_e$  part of the birth rates in

$$\sum_{e=0}^E \{\lambda q_e \Delta_e g(W) - W_e \Delta_e g(W - \mathbf{e}_e)\}.$$

Only the good indicators remain to be telescoped. The target parameters are still  $\lambda' = \lambda S_E$  and  $\mu_e = q_e/S_E$ , so  $\lambda'\sqrt{\mu_e \mu_f} = \lambda\sqrt{q_e q_f}$  exactly. No comparison of the good-site intensity with  $\lambda$  is needed, even if every site was removed. The independent vector  $Z$  stays in the outside count during each telescoping step, so the good-site dependency graph is unchanged. For good sites, apply the multivariate dependency-graph Stein argument [4, 15] to the *counts by mark*, not to counts by site and mark. On the first-difference telescoping terms each pair of types contributes its joint or product mass multiplied by  $|\Delta_{ef}g|$ . The product masses are  $p_e p_f$ . For local pairs use (C.1); for separated pairs use (C.2) after averaging. Decompose  $2^{\rho Q} = 1 + (2^{\rho Q} - 1)$  and sum the second part by  $R_2(N, Q)$ . The diagonal and all local pairs are bounded by  $O(NQp^2)$ .

Using the first bound in (C.3) sums the mark weights to at most one. Using the second sums them to  $\sum_{e,f} \sqrt{q_e q_f} \leq (1 + \sqrt{2})^2$ . Take the better of these two estimates. This proves (C.5) when  $E \geq 1$ . For  $E = 0$ , retain both signs during the same comparison, so there are two positive target coordinates. At fixed excesses the factor does not depend on the chosen signs; summing their probabilities before the relative-pair bound again gives  $R_2(N, Q)$ . The signed local factor is at most four, an absolute constant. Thus the result also covers  $E = 0$  without extending the literature statement to dimension one. Notice that the arithmetic argument is used at  $Q$ , rather than merely at  $L$ : after insertion of the directional weights, the unweighted folding identity for marks is not the inequality being summed.  $\square$

At  $Y = Y_N^\sharp$ , the first two arithmetic error sources in (C.5) are bounded by

$$\lambda e^{-V_N + o(\nu_N)} + N^{-1/2 + o(1)} \lambda, \quad \lambda(1 + \log^+(2\lambda)) e^{-V_N + o(\nu_N)}.$$

For clarity, before absorbing subpolynomial factors the profile term is

$$p^2 R_2(N, Q) \ll N^{-1/3 + \varepsilon} (\lambda^2 + \lambda 2^{E+2}).$$

If  $I + \log^+ \lambda = O(V_N)$  and  $E = O(V_N)$ , multiplication by  $e^I s(\lambda)$  leaves only an  $N^{o(1)}$  factor. Start with a smaller arithmetic  $\varepsilon$  to obtain the final remainder  $N^{-1/3 + \varepsilon}$  at any prescribed positive  $\varepsilon$ . This is the step that prevents an unrecorded exponential mark-cutoff loss.

For aggregation with signs retained, replace  $q_e$  by  $q_{e,s} = q_e/2$  and  $R_2(N, Q)$  by  $R_2^{\text{val}}(N, Q+1)$  in (C.5). The local factor is at most four and  $\sum_{e,s} \sqrt{q_{e,s}} = \sqrt{2}(1 + \sqrt{2})$ , both fixed constants. The full-value profile at  $Q+1$  vertices has the same envelope as the relative profile at  $Q$  rows. It therefore leaves the same polynomial remainder and the same one-factor budget. By contrast, the unweighted labelled argument sums signs and marks first and continues to use  $R_2(N, L)$  at the base length.

#### C.4 Removing the mark cutoff

Assume  $I + \log^+ \lambda \leq V_N - c\nu_N$ , and choose  $0 < c' < c'' < c$ . Set

$$E_N = \left\lceil \frac{\log^+ \lambda + c'' \nu_N}{\log 2} \right\rceil.$$

Then  $E_N = O(V_N) = o(\log N)$ , and the target tail has mass  $\mu_N = \lambda 2^{-E_N - 1} \leq e^{-c'' \nu_N}$ . The actual tail event is exactly a hit at length  $L + E_N + 1$ . The scalar hard-conditioning inequality at that shifted length gives

$$\mathbb{P}(\exists x, e > E_N : K_{x,e} = 1 \mid C) \leq \mu_N + O\left(e^I \mu_N (e^{-V_N + o(\nu_N)} + N^{-1/3 + \varepsilon})\right).$$

It is  $O(\mu_N)$  under the displayed budget. Couple each tail to zero, apply the finite theorem, and absorb  $\log(1 + \log^+ \lambda) = o(\nu_N)$ . This gives

$$d_{\text{TV}} \left( \mathcal{L}((X_e)_{e \geq 0} \mid C), \bigotimes_{e \geq 0} \text{Pois}(\lambda q_e) \right) \ll e^{-c' \nu_N} + N^{-1/3 + \varepsilon}.$$

For finite-mass sequences, tail summation is a measurable bijection, with inverse  $X_e = T_e - T_{e+1}$ . The entire threshold path has exactly the same total-variation distance.

### C.5 Keeping the spatial labels

For the labelled comparison no directional Stein factor is inserted. The identities

$$\sum_e p_e \leq p, \quad \sum_{e,f} \mathbb{P}(K_{x,e} K_{y,f} = 1) \leq \mathbb{P}(J_{x,L} J_{y,L} = 1)$$

fold the separated-pair contribution at the base length. Their signed versions hold after summing both signs. Thus the relation profile is  $R_2(N, L)$ , whereas the graph and deleted set use the maximal support.

**Free-cutoff bound and tails.** Let  $H = \log N$ ,  $w \in [V_N, \hat{V}_N]$ ,  $Y = \lfloor e^w \rfloor$  and  $\nu_w = H/w$ . Suppose  $C \in \mathcal{F}_Y$  has  $I = -\log \mathbb{P}(C) \geq 0$ , and  $\lambda = N2^{-L} \geq 1$  with  $\ell = \log \lambda$  and  $I + \ell = O(V_N)$ . Choose

$$E = \left\lceil \frac{I + \ell + w}{\log 2} \right\rceil, \quad Q = L + E + 1.$$

Since  $L = (H - \ell)/\log 2$ , the maximal number of values is

$$Q + 1 = \frac{H + I + w}{\log 2} + O(1) = \log_2 N + O(V_N).$$

Both the base and maximal lengths remain in a fixed positive logarithmic band, and  $Y > 2(Q+1)$  eventually, uniformly in these parameters. By [Theorem 5.8](#) and the uniform free-cutoff estimates, the finite field distance, multiplied by the information cost, is at most

$$C \left\{ e^{I+\ell-D(H/w)+o(\nu_w)} + e^{I+2\ell-w+o(\nu_w)} + e^I \lambda (1 + \lambda) N^{-1/3+\varepsilon/2+o(1)} \right\}. \quad (\text{C.6})$$

The weighted one-window term and the  $N^{-1+o(1)}$  part of the edge bound fit the polynomial term. Constants in the little-oh terms are uniform for  $w$  in the displayed interval and for the enlarged lengths. A fixed multiplicative enlargement of this interval is also permitted by [proposition B.1](#).

An excess greater than  $E$  is exactly a threshold hit at  $L + E + 1$ . The unconditional first moment gives actual conditional tail cost at most

$$e^I \lambda 2^{-E-1} (1 + N^{-1/2+o(1)}) = O(e^{-w}),$$

and the target tail has no larger cost. This step uses conditioning only as the bound  $\mathbb{P}(H | C) \leq e^I \mathbb{P}(H)$ ; no conditioning theorem at a different cutoff is invoked. Coupling both tails to zero adds  $O(e^{-w})$  to (C.6). The full site–sign–excess space is countable, with finite expected total mass, so these tail couplings exhaust its coordinates.

In the applications below,  $I + \ell \leq w \leq \hat{V}_N$ . One may instead use the common truncation  $E = 3\lceil \hat{V}_N / \log 2 \rceil$ , for which  $e^I \lambda 2^{-E-1} \leq \frac{1}{2} e^{-2\hat{V}_N}$ . It still has  $Q + 1 = \log_2 N + O(V_N)$ , so the finite comparison is unchanged and both tails are  $O(e^{-2\hat{V}_N})$ . This is the truncation used in the proof of [proposition 6.2](#); the sharper choice above is not needed for that conclusion.

**The information-dependent crossing.** For  $0 \leq I \leq V_N$ , set  $F_I(w) = 2D(H/w) - w - I$ . The function  $D$  is continuous and strictly increasing on the larger saddle branch. Since  $H/w$  strictly decreases with  $w$ ,  $F_I$  is continuous and strictly decreasing. Its endpoint values on  $[V_N, \hat{V}_N]$  are  $V_N - I$  and  $-I$ . There is therefore exactly one root  $w_N(I)$ , including the endpoints for  $I = V_N$  and  $I = 0$ . The first branch of  $g_{N,I}(w) = \min\{D(H/w) - I, (w - I)/2\}$  decreases strictly; the second increases strictly. The root is their unique maximum on the interval. It

obeys

$$D(H/w_N(I)) = \frac{w_N(I) + I}{2}, \quad \ell_{\text{lab},N}(I) = D(H/w_N(I)) - I.$$

If  $\ell \leq \ell_{\text{lab},N}(I) - c\nu_w$ , the first two terms in (C.6) are respectively  $e^{-c\nu_w + o(\nu_w)}$  and  $e^{-2c\nu_w + o(\nu_w)}$ . The polynomial multiplier is  $N^{o(1)}$  because  $I + 2\ell \leq w_N(I) = O(V_N)$ . Also  $w/\nu_w = w^2/H \asymp \log H \rightarrow \infty$ , so  $e^{-w}$  is smaller than any fixed  $e^{-c'\nu_w}$ . This proves [proposition 6.2](#). The hypotheses are vacuous at  $I = V_N$  for  $\lambda \geq 1$ , since then the allowed leading logarithmic intensity is zero and the margin is strictly negative.

For  $I > 0$ , the root satisfies  $w_N(I) < \hat{V}_N$ . Hence  $2D(H/w_N(I)) > 2D(H/\hat{V}_N) = \hat{V}_N$ , and its defining equation gives  $w_N(I) + I > \hat{V}_N$ . Thus  $w_N(I) > \hat{V}_N - I$ , giving the claimed strict improvement of the leading budget over  $\hat{V}_N/2 - I$  without differentiating the implicitly defined root. Uniformly for  $w = tV_N$ ,  $1 \leq t \leq \hat{V}_N/V_N$ , the expansion  $D(\nu) = \nu u(\nu)(1 + O(1/u(\nu)))$  gives  $D(H/w)/V_N = t^{-1} + o(1)$ . The saddle equation consequently yields  $t^2 + \theta t - 2 = 0$  when  $I/V_N \rightarrow \theta$ . The positive root is  $(\sqrt{\theta^2 + 8} - \theta)/2$ ; subtracting  $\theta$  and dividing by two proves (6.8).

**A fixed event and a constrained cutoff.** Suppose  $C \in \mathcal{F}_{[e^{w_0}]}$  for a given  $w_0 \in [V_N, \hat{V}_N]$ . The prime sigma-fields increase with the cutoff. On the certified admissible interval  $[w_0, \hat{V}_N]$ ,  $g_{N,I}$  increases up to  $w_N(I)$  and decreases afterwards; hence its maximum is at  $\max\{w_0, w_N(I)\}$ . For any admissible  $w$  the two inequalities

$$I + \ell \leq D(H/w) - cH/w, \quad I + 2\ell \leq w - 2cH/w$$

are sufficient. In particular no event measurable only above the proposed cutoff is retained by an algebraic optimization. All cutoffs here are deterministic functions of the event and its probability, not of the realized prime environment. This is an optimization of the proved labelled bounds, not an optimality assertion about the law.

## C.6 Macroscopic transport and containment

Let  $L_M = \lfloor \log_2 M \rfloor - a_M$ ,  $a_M = o(\log M)$ , and  $\lambda = M2^{-L_M}$ . The base-contained interior starts are  $2 \leq x \leq M - L_M + 1$ ; their number is  $M - L_M$ . First retain  $\lceil M^{1/2} \rceil \leq x \leq M - L_M + 1$ . The conditional ranks and graph construction are local and unchanged. The one-window defect masses, summed on dyadic slices, are  $M^{1/2+o(1)}$ ; the edge estimate uses the upper height  $M$ ; and the separated-pair mass is bounded by the macroscopic profile (3.4). Thus the labelled finite comparison is

$$\ll \lambda(1 + \lambda)(e^{-V_M + o(\nu_M)} + M^{-1/3+\epsilon}).$$

Restoring the deep starts costs the bound (7.7),

$$O\left(\lambda M^{-1/2} + \exp(-c_0 \log M / \log \log M)\right).$$

The target deletion costs only  $O(\lambda M^{-1/2})$ . Conditioning on a compatible event in the chosen prime field multiplies the averaged error by at most  $e^I$ .

Choose  $E = \lceil (I + V_M + \log(2 + \lambda)) / \log 2 \rceil$  in the hard labelled regime. The exact mark-tail identity and the global scalar bound control the real tail. Besides the intensity-linear tail term, the scalar bound has the separate deep-start error just displayed; that error is already present in the full-field inequality. In particular it must not be silently dropped from a finite-prefix tail calculation. Both tails are absorbed by (7.14). The free-cutoff proof is identical with  $V_M$  replaced by the optimized cutoff in its two leading terms.

For aggregated counts apply Theorem C.1 on the retained macroscopic set, with  $R_{2,\delta}(M, Q)$  in place of  $R_2(N, Q)$ . The same geometric square-root sum applies. Under  $I + \log^+ \lambda \leq V_M - c\nu_M$ , the deep-start correction times  $e^I$  is smaller than every fixed  $e^{-c'\nu_M}$ , and the aggregate mark cutoff above removes both tails. This proves the macroscopic one-factor staircase without an additional marked arithmetic input.

At exact centered level  $r$  the target mean is precisely  $(M - L_M)2^{-b_M-r-1}$ , not  $M2^{-b_M-r-1}$ . All these fields retain the full rightward mark of a base-contained start. They are not the list of blocks censored at the right end of the finite sequence. The containment diagram in the article distinguishes the two objects.

## C.7 Deterministic masks and diffuse limits

A deterministic restriction of a lattice field contracts total variation. At bounded intensity, if  $N^{-1} \sum_{x \in A_N} \delta_{x/N} \Rightarrow \mu$ , the restricted marked field therefore converges weakly to  $\text{PPP}(\lambda\mu \otimes \nu_{\text{geo}})$ , along a subsequence  $N2^{-L} \rightarrow \lambda$ . The same statement holds for the macroscopic field, with scaling by  $M$ . It follows directly from the target Laplace functional and weak convergence of the deterministic empirical measures. There is no universal numerical rate for this last deterministic step without a quantitative assumption on the mask or the test function.

Nor is the diffuse limit a total-variation target. The actual field is supported on its lattice, while the diffuse target has positive probability  $1 - e^{-\lambda\mu(\mathbb{R})}$  of a nonempty configuration outside that lattice. The approximation in total variation always takes place *before* passing from the grid to the continuum.

## D Conditional laws and the Poisson target

The arithmetic theorems provide total-variation errors. This appendix records exactly when those errors are small enough to transfer a local probability, a conditional state, or a Gaussian limit. Properties of the independent Poisson target are not additional arithmetic results.

### D.1 Scalar Gaussian and local estimates

Let  $P_\lambda \sim \text{Pois}(\lambda)$  and let  $\epsilon_N$  denote any proved conditional scalar total-variation bound in the article. If  $\lambda \rightarrow \infty$ ,

$$\sup_t \left| \mathbb{P} \left( \frac{Z - \lambda}{\sqrt{\lambda}} \leq t \mid C \right) - \Phi(t) \right| \leq \epsilon_N + C\lambda^{-1/2}.$$

For example, write  $P_\lambda$  as a sum of  $\lceil \lambda \rceil$  independent Poisson variables of mean at most one and use the classical Berry bound [5]. Their third absolute centered moments are  $O(\lambda/\lceil \lambda \rceil)$ , which gives the displayed scale. The normalization uses the deterministic comparison mean  $\lambda$ ; replacing it by an actual conditional mean requires a separate moment estimate and is not inferred from total variation alone.

For every integer  $n \geq 1$  and every  $\lambda > 0$ , one has the exact form

$$\mathbb{P}(P_\lambda = n) = \frac{e^{-\delta_n}}{\sqrt{2\pi n}} \exp[-\lambda h(n/\lambda)], \quad 0 \leq \delta_n \leq \frac{1}{12n}. \quad (\text{D.1})$$

Here  $h(u) = u \log u - u + 1$ , and  $\delta_n$  is independent of  $\lambda$ . To check the remainder, set

$$\delta_n = \log(n!) - (n + \tfrac{1}{2}) \log n + n - \tfrac{1}{2} \log(2\pi).$$

Stirling's limit gives  $\delta_n \rightarrow 0$ . With  $t = (2n+1)^{-1}$ ,

$$\begin{aligned} \delta_n - \delta_{n+1} &= (n + \tfrac{1}{2}) \log(1 + 1/n) - 1 = \sum_{j \geq 1} \frac{t^{2j}}{2j+1}, \\ 0 \leq \delta_n - \delta_{n+1} &\leq \frac{t^2}{3(1-t^2)} = \frac{1}{12n(n+1)}. \end{aligned}$$

Telescoping to infinity proves the stated bounds. Substitution of the factorial identity into  $e^{-\lambda} \lambda^n / n!$  proves (D.1); moreover  $0 \leq 1 - e^{-\delta_n} \leq 1/(12n)$ , uniformly in the intensity. For  $n = 0$  the mass is exactly  $e^{-\lambda}$  and the square-root formula is not used.

Thus relative transfer at  $n$  requires  $\epsilon_N / \mathbb{P}(P_\lambda = n) \rightarrow 0$ . For the hard scalar rate a sufficient condition is

$$I + \log^+ \lambda + \tfrac{1}{2} \log n + \lambda h(n/\lambda) \leq V_N - c\nu_N,$$

and for the soft rate a sufficient condition is

$$2I + \log^+ \lambda + \log n + 2\lambda h(n/\lambda) \leq \widehat{V}_N - c\widehat{\nu}_N.$$

In both cases the event must be measurable at the corresponding prime cutoff. The logarithms of the reciprocal target masses are then  $O(V_N)$  or  $O(\widehat{V}_N)$ , so division by that mass does not destroy the polynomial remainder.

In the central range  $n = \lambda + t\sqrt{\lambda} + O(1)$ ,  $|t| = o(\lambda^{1/6})$ , Taylor expansion gives  $\lambda h(n/\lambda) = t^2/2 + o(1)$  and  $\log n = \log \lambda + o(1)$ . The respective central local budgets are therefore

$$I + \tfrac{3}{2} \log \lambda + \tfrac{1}{2} t^2 \leq V_N - c\nu_N, \quad 2I + 2 \log \lambda + t^2 \leq \widehat{V}_N - c\widehat{\nu}_N.$$

A global scalar approximation at its upper intensity endpoint does not by itself imply a relative local approximation there.

For completeness, the analogous relative upper-tail transfer has the additional denominator  $\mathbb{P}(P_\lambda \geq \lceil \lambda + t\sqrt{\lambda} \rceil)$ . Stirling's formula and summation of successive Poisson masses yield the usual Gaussian moderate-tail comparison for  $0 \leq t = o(\lambda^{1/6})$ . Equivalently, expanding the exponent uniformly on the contributing  $O(\sqrt{\lambda}/(1+t))$  scale gives relative error  $O((1+t^3)/\sqrt{\lambda})$ ; the exponentially smaller outer part of the sum is bounded by the ratio of successive masses. Since  $1 - \Phi(t) \asymp e^{-t^2/2}/(1+t)$  for  $t \geq 0$ , a sufficient hard arithmetic condition is

$$I + \log^+ \lambda + \tfrac{1}{2} t^2 + \log(1+t) \leq V_N - c\nu_N.$$

This condition must be combined with the intrinsic Poisson restriction  $t = o(\lambda^{1/6})$ . It is not a claim about larger deviation scales.

## D.2 Resolving an entire conditional path

Let  $\epsilon^{\text{path}}$  be the aggregate staircase error based at the present threshold. The sharpened conditioning inequality (6.9) gives a full future-path error at most  $\epsilon^{\text{path}}/p_\lambda(n)$ , where  $p_\lambda(n) = \mathbb{P}(P_\lambda = n) > \epsilon^{\text{path}}$ . The same assertion holds jointly with any fixed available lower segment. Forward transitions are binomial thinning; backward transitions add independent Poisson increments.

Before division, the aggregate proof gives

$$\epsilon^{\text{path}} \ll e^I \lambda (1 + \log^+(2\lambda)) e^{-V_N + o(\nu_N)} + N^{-1/3+\varepsilon},$$

provided the information, log-intensity and mark cutoff are  $O(V_N)$ . Combining this with (D.1) proves the budget (6.12). Its hypotheses also validate that use of the common logarithmic band. Indeed, for  $n \geq 2$ , if  $\lambda \geq 1$  the budget bounds  $\log \lambda$  by  $O(V_N)$ . If  $\lambda < 1$ , use

$$\lambda h(n/\lambda) = n \log n - n + \lambda - n \log \lambda$$

to conclude  $|\log \lambda| = O(V_N)$ . Thus the base offset, information cost and chosen mark cutoff are all  $O(V_N) = o(\log N)$ . The atom's reciprocal is  $\exp(O(V_N)) = N^{o(1)}$ , which can be absorbed by first choosing a smaller arithmetic  $\varepsilon$ . For a fixed present state at a fixed positive limiting intensity the atom stays bounded away from zero, and no extra logarithmic cost is needed.

The condition  $p_\lambda(n) > \epsilon^{\text{path}}$  only makes the conditional law well-defined and gives a bound. Convergence requires  $\epsilon^{\text{path}}/p_\lambda(n) \rightarrow 0$ . Nothing here supplies a single finite-intensity comparison for the entire reverse path: its target mass diverges toward thresholds  $m \rightarrow -\infty$ .

### D.3 Quenched transfer along geometric scales

The proof of Corollary 6.5 gives the single-length Markov–Borel–Cantelli argument. Its uniform extension uses the bound

$$e^{-(c-\beta)\nu_{N_k} + o(\nu_{N_k})} + e^{\beta\nu_{N_k}} N_k^{-1/3+\varepsilon}, \quad 0 < \beta < c,$$

for the exceptional probability on  $N_k \asymp q^k$ ,  $q > 1$ . Since  $\nu_{N_k} \asymp \sqrt{k/\log k}$ , multiplication by  $O(\log N_k) = O(k)$  preserves summability. A single union bound therefore covers every integer length in a fixed logarithmic band, provided the error margin is uniform. The resulting almost-sure statement concerns these conditional laws at individual scales; it requires no independence between scales and asserts no canonical coupling over all scales.

### D.4 Exact target consequences

For a phase  $\vartheta \in [0, 1]$ , let  $P_r \sim \text{Pois}(2^{\vartheta-r-1})$  independently for  $r \in \mathbb{Z}$ , and  $S_m = \sum_{r \geq m} P_r$ . Each tail sum is finite almost surely. Then

$$S_m \sim \text{Pois}(2^{\vartheta-m}), \quad \text{Cov}(S_m, S_n) = 2^{\vartheta-\max(m,n)}.$$

Splitting  $S_m = P_m + S_{m+1}$  gives the exact forward thinning and backward immigration kernels in the article. For  $s \geq 0$  the joint probability generating function is

$$\mathbb{E} \prod_{j=0}^s z_j^{S_{m+j}} = \exp \left[ \sum_{r=m}^{m+s-1} 2^{\vartheta-r-1} \left( \prod_{j=0}^{r-m} z_j - 1 \right) + 2^{\vartheta-m-s} \left( \prod_{j=0}^s z_j - 1 \right) \right].$$

Conditionally on  $S_m = n$ , the  $n$  points have independent uniform positions and independent geometric overshoots  $\mathbb{P}(E = e) = 2^{-e-1}$ . For lattice targets the positions are uniform on the corresponding finite grid. Conditioning the critical prelimit field on a fixed count transfers this assertion by the sharp conditioning lemma; only the final grid-to-uniform passage is weak convergence.

The complete centered target is locally finite on  $[1, 2] \times \mathbb{Z}$ , although it has infinite total mass toward  $r = -\infty$ . For every nonnegative continuous compactly supported  $g$  put

$$\Psi_\vartheta(g) = \exp \left[ - \int_1^2 \sum_{r \in \mathbb{Z}} 2^{\vartheta-r-1} (1 - e^{-g(t,r)}) dt \right].$$

Compact support meets finitely many levels. At those levels the lattice comparison and a Riemann sum give

$$\mathbb{E}(e^{-\langle g, \mathcal{E}_N \rangle} \mid C_N) = \Psi_{\vartheta_N}(g) + o_g(1)$$

whenever  $I_N \leq V_N - c\nu_N$ . A convergent phase subsequence is needed only to replace this moving deterministic target by a single limit law.

For a base intensity  $\lambda_N \rightarrow \infty$ , the exact maximum identity in the article, applied at  $m_N = \lfloor \log_2 \lambda_N \rfloor + j$ , gives, along  $\{\log_2 \lambda_N\} \rightarrow \vartheta$ ,

$$\mathbb{P}(M_{N,L} - \lfloor \log_2 \lambda_N \rfloor \leq j) \longrightarrow e^{-2^{\vartheta-j-1}}.$$

This is a lattice extreme law. The identity uses the scalar comparison at the shifted critical length, not an unproved uniform law for every large marked field.

Finally, the Poisson–Gaussian bridge follows by splitting the independent levels at a fixed critical level  $r_0$ . Below it, the normalized Poisson sums satisfy a multivariate central limit theorem with covariance  $2^{-\max(j,k)}$ . The centered contribution of the levels at or above  $r_0$  has variance  $O(\Lambda_N^{-1})$  after normalization, whereas those levels retain their Poisson law and are independent of the lower part. This proves joint independence in the limit. No continuous-time embedding or phase-removing auxiliary randomization is used in this arithmetic argument.

## E The microscopic boundary and relative factorization

The exact border cylinder and microscopic non-vacancy are different objects. The first has probability  $2^{-\pi(L)}$  without an error term; the second also allows an interior start. This appendix proves that the entire interior first moment is exponentially smaller than the border probability. A union bound is then sufficient; no microscopic overlap expansion is needed.

### E.1 Valuation matrices and the transition range

Put  $B = L + 1$ ,  $V_x = \{x - 1, \dots, x + L - 1\}$ , and let  $M_x$  have rows  $v(n) = (v_p(n) \bmod 2)_p$  for  $n \in V_x$ . The start matrix is  $A_x = D_x M_x$ , where  $D_x$  is the edge-difference matrix of a tree on  $V_x$ . Its kernel consists exactly of the constant vectors. Consequently

$$\text{rank } A_x \geq \text{rank } M_x - 1, \quad \mathbb{P}(J_{x,L} = 1) \leq 2^{-\text{rank } A_x}. \quad (\text{E.1})$$

For  $2 \leq x \leq L - 1$  a start is impossible: both  $2x$  and  $2x - 2$  lie inside the putative constant run, and complete multiplicativity would force  $f(2) = 1$  and  $f(2) = -1$  simultaneously.

**Lemma E.1** (Transition identity minor). *Uniformly for  $x \in \{B - 1, B, B + 1\}$ ,  $\text{rank } A_x \geq (2 - o(1))\pi(B)$ .*

*Proof.* Write  $a = x - 1 \in \{B - 2, B - 1, B\}$  and  $Y_0 = 2\sqrt{B}$ . For each prime  $Y_0 < p \leq B$  choose  $m_p = p \lceil a/p \rceil \in [a, a + B - 1]$ . The cofactor satisfies  $\lceil a/p \rceil \leq B/p + 1 < \sqrt{B}/2 + 1 < Y_0 < p$  for large  $B$ . Thus  $m_p$  has valuation one at  $p$  and no other prime factor above  $Y_0$ . Each prime  $B < q \leq a + B - 1$  occurs as its own vertex. On these rows and columns the valuation matrix has an identity minor of size  $\pi(a + B - 1) - \pi(Y_0) = (2 - o(1))\pi(B)$ . Apply (E.1).  $\square$

## E.2 The uniform prime-divisor input

The precise literature input is

$$\forall \varepsilon > 0 \exists B_0(\varepsilon) \forall B \geq B_0(\varepsilon) \forall n > B : \quad \omega \left( \prod_{j=0}^{B-1} (n+j) \right) \geq (2 - \varepsilon) \pi(B). \quad (\text{E.2})$$

It is equation (14) of Laishram–Shorey [13]. Its uniformity is important. For  $n > 17B/12$ , their Theorem 2 gives at least  $\pi(2B)$  prime divisors, apart from a fixed exceptional pair. For  $B < n \leq 17B/12$ , every prime at most  $B$  divides the block product, and each prime in  $[n, n+B-1]$  is an additional divisor. Hence the number is at least

$$\pi(B) + \pi(n+B-1) - \pi(n-1) = (2 - o(1))\pi(B),$$

uniformly in that bounded ratio, by the prime number theorem. This verifies the quantifier order in (E.2).

Suppose now that  $B+2 \leq x \leq B^2+2$ . Every vertex is at most  $B^2+B$ . Equation (E.2) supplies  $t_x \geq (1 - o(1))\pi(B)$  primes  $q > B$  in the block. Each has valuation one, since  $q^2 \geq (B+1)^2 > B^2+B$ ; it occurs at one vertex only, since  $q > B-1$ ; and two such primes cannot occupy the same vertex, since their product exceeds  $B^2+B$ . Their columns span a coordinate subspace  $U_x$  of dimension  $t_x$ . Call its rows the private rows.

## E.3 A global count of even valuations

Fix an integer  $K \geq 2$  and use the medium primes  $B/K < p \leq B$ . Let  $E_x$  be their total odd-valuation incidence mass in  $V_x$ . We claim

$$E_x \geq (H_K - 1 - o(1))\pi(B), \quad H_K = \sum_{j=1}^K \frac{1}{j}. \quad (\text{E.3})$$

The block contains at least  $\lfloor B/p \rfloor$  multiples of  $p$ . An incidence of positive even valuation is among those with  $n = ap^2 \leq B^2+B$ . Thus  $a \leq K^2+1$  for sufficiently large  $B$ . For fixed  $a$ , successive distinct primes  $p, q > B/K$  give values  $ap^2, aq^2$  separated by more than  $2aB/K$ ; only  $1 + K/(2a)$  such values can lie in an interval of length  $B$ . Summing over the fixed number of possible  $a$  shows that the total number of square divisibilities is  $O_K(1)$ , uniformly in  $x$ . Subtracting all of them is a permissible overcount of even valuations.

The main incidence count can be evaluated without rounding every summand:

$$\begin{aligned} \sum_{B/K < p \leq B} \left\lfloor \frac{B}{p} \right\rfloor &= \sum_{j=1}^{K-1} (\pi(B/j) - \pi(B/K)) \\ &= (H_{K-1} - (K-1)/K + o(1))\pi(B) \\ &= (H_K - 1 + o(1))\pi(B). \end{aligned}$$

This proves (E.3). The improvement over a prime-by-prime loss is essential: subtracting one for each prime would lose a positive proportion of  $\pi(B)$ .

## E.4 Rank of a simple incidence matrix

**Lemma E.2** (Simple incidence rank). *Let  $H$  be a finite binary matrix. Suppose every row has weight at most two, every column has weight at most  $K$ , and no pair of columns occurs as the*

support of two weight-two rows. If  $E$  is the number of entries equal to one, then

$$\text{rank } H \geq \frac{E}{K+1}.$$

*Proof.* Remove zero columns and view a weight-two row as an edge between its columns, and a weight-one row as a pin. In a connected component with  $r$  active columns, an unpinned component has rank  $r - 1$  and a pinned component has rank  $r$ . The first assertion follows because the edge vectors of a connected graph span the even-parity subspace; one pin then spans its missing direction.

A pinned component has incidence mass at most  $Kr$ , which is at most  $(K+1)$  times its rank. An unpinned component has  $r \geq 2$ ; simplicity and the degree bound give incidence mass at most  $\min\{r(r-1), Kr\}$ . If  $r \leq K+1$  use the first bound, and if  $r \geq K+1$  use the second. In either case the mass is at most  $(K+1)(r-1)$ . Sum over the components.  $\square$

Delete the private rows from the medium-prime columns of  $M_x$ , and call the resulting matrix  $H_x$ . For fixed  $K$  and sufficiently large  $B$ , a private row carries at most one medium prime: otherwise a vertex would exceed  $B(B/K)^2 > B^2 + B$ . Thus deletion removes at most  $t_x$  incidences. A remaining row carries at most two medium primes, since  $(B/K)^3 > B^2 + B$ . A column has weight at most  $K$ . Finally two weight-two rows with the same support  $p, q$  would be two multiples of  $pq > (B/K)^2 > B$  inside one block, which is impossible. All hypotheses of Lemma E.2 are therefore met. Modulo  $U_x$ , a medium column is represented by its restriction to the remaining rows, so

$$\begin{aligned} \text{rank } A_x &\geq t_x + \frac{(E_x - t_x)_+}{K+1} - 1 \\ &\geq \left(1 + \frac{H_K - 2}{K+1} - o(1)\right) \pi(B). \end{aligned} \tag{E.4}$$

For the last step, if  $E_x \leq t_x$  use  $t_x \geq E_x$ ; otherwise use  $t_x \geq (1 - o(1))\pi(B)$  and (E.3). The parameter can be chosen exactly within this family. Set  $\delta_K = (H_K - 2)/(K+1)$ . Then

$$\delta_{K+1} - \delta_K = \frac{3 - H_K}{(K+1)(K+2)}.$$

Since  $H_{10} < 3 < H_{11}$  and  $H_K$  is strictly increasing, the unique maximizer over integers  $K \geq 2$  is  $K = 11$ . Consequently we use

$$H_{11} = \frac{83711}{27720}, \quad \delta_* := \frac{H_{11} - 2}{12} = \frac{28271}{332640} > \frac{1}{12}.$$

This optimizes only the displayed fixed-cutoff family of bounds, not the rank surplus among all possible methods.

The local product comparisons hold for  $B \geq 11^3 + 1 = 1332$ :  $(B/11)^3 > B^2 + B$ ,  $B^3/121 > B^2 + B$  and  $(B/11)^2 > B$ . In the square-divisibility count one may then use  $a \leq 122$  and at most seven values per  $a$ , giving the safe bound 854. These elementary checks do *not* make 1332 a threshold for the complete asymptotic theorem: the prime-divisor and prime number theorem thresholds still depend on the desired error margin.

## E.5 Post-quadratic starts and summation

**Lemma E.3** (Post-quadratic defect maximum). *There is an absolute constant  $\theta_0$  such that, for all sufficiently large  $B$  and every  $x > B^2 + 2$ , the number  $m_x$  of  $B$ -defective vertices in  $V_x$*

is at most  $B - g_L$ , where

$$g_L = \frac{B}{\log B} (\log \log B - \log \log \log B - \theta_0), \quad g_L / \pi(B) \longrightarrow \infty.$$

In particular  $\mathbb{P}(J_{x,L} = 1) \leq 2^{-g_L}$ .

*Proof.* We use the Balasubramanian–Shorey exclusion [3] in the form recorded by Shorey [16, pp. 55–56, equation (15) with exponent 2 and the subsequent square-case statement]. It has an absolute constant  $\theta_0$  and, for every fixed  $\varepsilon > 0$ , a threshold  $k_0(\varepsilon)$  such that, for  $k \geq k_0(\varepsilon)$ ,

$$\prod_{i=1}^t (z + d'_i) = by^2, \quad 0 \leq d'_1 < \cdots < d'_t < k, \quad b > 0, \quad P^+(b) \leq k, \quad z > e^{1-\theta_0+\varepsilon} \frac{k \log k}{\log \log k}$$

forces  $t < \mu_k(\theta_0)$ , where

$$\mu_k(\theta) = k \left( 1 - \frac{\log \log k}{\log k} + \frac{\log \log \log k}{\log k} + \frac{\theta}{\log k} \right).$$

The smoothness condition on  $b$  is part of (15); it is not a size bound on  $b$ . All thresholds are uniform in  $z, b, y$  and the shifts. To obtain the form used here, first fix  $\varepsilon = 1$ . Increase the absolute threshold until

$$k^2 > e^{2-\theta_0} \frac{k \log k}{\log \log k}.$$

For  $m > k^2$  and  $1 \leq d_1 < \cdots < d_t \leq k$ , set  $z = m + 1$  and  $d'_i = d_i - 1$ . These substitutions preserve every factor  $m + d_i = z + d'_i$ , put all shifts in  $[0, k)$ , and satisfy the required height bound. Hence the product equation forces  $t < \mu_k(\theta_0)$ . Write  $V_x = \{m + 1, \dots, m + B\}$  with  $m = x - 2 > B^2$ . The product of its  $B$ -defective vertices is a square times a  $B$ -smooth integer. There is no extra upper bound on that smooth coefficient in the imported statement. Its contrapositive gives  $m_x < \mu_B(\theta_0) = B - g_L$  beyond the fixed absolute threshold. The affine defect bound is  $\mathbb{P}(J_{x,L} = 1) \leq 2^{-L} 2^{\max(m_x-1, 0)}$ , which is at most  $2^{-g_L}$  for large  $B$ .  $\square$

Partition the starts  $2 \leq x \leq 2L^2$  into the forbidden early range, the three transition starts,  $B + 2 \leq x \leq B^2 + 2$ , and the remaining post-quadratic starts. Their total probabilities are respectively

$$0, \quad 3 \cdot 2^{-(2-o(1))\pi(B)}, \quad O(B^2) 2^{-(1+\delta_*-o(1))\pi(B)}, \quad O(L^2) 2^{-g_L}.$$

Since  $\log B = o(\pi(B))$  and  $\pi(B) = \pi(L) + O(1)$ , their sum is at most  $2^{-(1+\delta_*-o(1))\pi(L)}$ . Adding the exact border event proves

$$q_L = 2^{-\pi(L)} (1 + O_{\delta_0}(2^{-\delta_0 \pi(L)})) \quad (0 < \delta_0 < 28271/332640),$$

including the displayed choice  $\delta_0 = 1/12$ . The same interior sum divided by  $q_L$  bounds the conditional probability that the exact border is not the unique microscopic start.

The cutoff-invariance and mesoscopic estimates use one further observation. At heights above  $2L^2$ , two  $B$ -defective vertices at distance at most  $L$  have distinct squarefree  $B$ -smooth parts; equal parts would place consecutive squares too far apart. The number of choices of these parts is  $\exp(O(L/\log L))$ , and the uniform Pell bound gives the same type of bound on the number of corresponding windows in the entire admissible population up to twice the ambient height. Multiplication by  $2^{-g_L}$  remains  $o(2^{-\pi(L)})$ , without an exterior height-slice factor. Starts with at most one defect cost only their number times  $2^{-L}$ . This proves the bounds in the article

with cutoff  $K \leq 2^L$ , using  $2^L$  as an artificial upper ambient scale; it does not require a dyadic lower endpoint comparable to that artificial scale.

## E.6 The prime clock

Section 7.6 of the article proves the exact prime-clock law and explains why integer overshoots need not be tight. We record here the conventions needed for the conditional comparisons. Let  $J_\partial = \min\{j : f(p_j) = -1\}$ . On  $C_L = 1$  and the full-measure event  $J_\partial < \infty$ , put  $G_L = J_\partial - \pi(L) - 1$ ; assign  $G_L = 0$  elsewhere. Then

$$\mathbb{P}(G_L = k \mid C_L = 1) = 2^{-k-1}, \quad \mathbb{P}(\ell_\partial - L \geq h \mid C_L = 1) = 2^{-[\pi(L+h) - \pi(L)]}.$$

Since  $\{C_L = 1\} \subseteq \{B_L^{\text{mic}} > 0\}$ , the conditional law of  $G_L$  given microscopic non-vacancy is a mixture whose  $C_L = 1$  component is exactly geometric. Its distance from that law is at most  $\mathbb{P}(C_L = 0 \mid B_L^{\text{mic}} > 0) = O(2^{-\delta_0 \pi(L)})$ . The next subsection constructs the finite-cylinder representative used to carry this mark through the stable product comparison.

## E.7 Relative bulk estimates and affine conditioning

Let  $\lambda = M2^{-L} \rightarrow 0$ ,  $L = O(\log M)$  and  $U = [\lceil M^\delta \rceil, M - L + 1] \cap \mathbb{Z}$ . Fix  $\delta$  and an upper bound for  $L/\log M$ . With  $Y = Y_M^\sharp$ , truncate the signed exact marks at  $E$  and put  $Q = L + E + 1$ . The averaged conditional process error is at most

$$C \left[ p\mathcal{M}_B + p|D_Y(Q)| + p^2(MQ + E_Y(Q) + R_{2,\delta}(M, L)) \right].$$

The local ranks are those in Appendix C. Summing signs and exact marks before the separated-pair estimate leaves the relation sum at the base length  $L$ ; only the bad supports and shared-prime graph use  $Q$ . After division by  $Mp = \lambda$ , the contributions are

| Source                       | Normalized contribution                    |
|------------------------------|--------------------------------------------|
| One-window defects           | $M^{-1/2+o(1)}$                            |
| Bad maximal supports         | $e^{-V_M+o(\nu_M)}$                        |
| Diagonal and local pairs     | $pQ$                                       |
| Shared-prime edges           | $\lambda(e^{-V_M+o(\nu_M)} + M^{-1+o(1)})$ |
| Base-length relations        | $M^{-1/3+\varepsilon}(1 + \lambda)$        |
| Actual and target mark tails | $O(2^{-E-1})(1 + M^{-1/2+o(1)})$           |

For the last row, the exact tail event is a start at length  $L + E + 1$ ; the masked first-moment estimate at that length gives the displayed bound after averaging the conditional tail probability. Choose  $E = \lceil V_M/\log 2 \rceil$ . Then  $Q = O(\log M)$ ,  $pQ = O(\log M/M)$  for large  $M$ , and every row tends to zero. This proves the signed, marked relative error in (7.23), not merely its unmarked projection.

The microscopic vector is measurable at the same prime cutoff because  $2L^2 + L < Y$  for large  $M$ . For a fixed  $K$ , define  $G_{L,K}$  from the first  $K$  prime signs after  $L$ : count consecutive positive signs, capped at  $K$ , and put the value zero off  $C_L = 1$ . This finite-cylinder variable agrees almost surely with  $G_L \wedge K$ ; this makes measurability on the finite prime sigma-field explicit, independently of the convention on the null nonterminating event. Bertrand's postulate bounds the  $K$ th prime after  $L$  by  $2^K L < Y$  eventually, for fixed  $K$ . The stable lift retains  $(\mathbf{B}_L, G_{L,K})$  jointly with the full signed marked bulk field, at cost  $\Delta_M = o(\lambda)$ .

Write  $\alpha = 2^{-\pi(L)}$ ,  $b = |U|p$ , and let  $u_M$  and  $v_M$  denote the interior microscopic first moment and the mesoscopic hit probability. The source bounds give  $u_M + v_M = o(\alpha + b)$ . The one-hit calculation in the proof of Theorem 7.11 gives sparse-target mass  $e^{-b}[\alpha + (1 - \alpha)b]$  and a change

of mixture weights at most  $b$  relative to (7.30). The conditional bulk mark, sign and position retain their full laws; only the boundary clock is truncated. Simultaneous sources and multiple bulk points have probabilities at most  $\alpha b$  and  $b^2/2$ . Since the sparse-event mass is asymptotic to  $\alpha + b$ , normalizing the source subprobability measures gives

$$d_{\text{TV}}(\mathcal{L}(\Gamma_{M,L}^{[K]} \mid R_M \geq L), Q_{M,L}^{[K]}) \ll \frac{\Delta_M + u_M + v_M + \alpha b + b^2}{\alpha + b} + b.$$

The additional  $b$  is already absorbed by the last two numerator terms. For the actual first-hit law the discarded boundary tail is at most  $2^{-K-1}$ : its unnormalized mass is at most  $\alpha 2^{-K-1}$  and the hit probability is at least  $\alpha$ . The target tail is also at most  $2^{-K-1}$ . First let  $M \rightarrow \infty$ , then let  $K \rightarrow \infty$ . No lower bound on either mixture weight, convergence of their ratio, or exact finite-size independence is required. The grid-to-Lebesgue passage is a separate weak limit.

Now let  $A = \{GX = b\}$  be a compatible affine cylinder on the exposed prime signs. Write  $r = \text{rank } G$ ,  $I = r \log 2$ , and  $U_L = \text{span}\{e_p : p \leq L\}$ . If the stack with  $C_L$  is compatible, its rank is  $r + \pi(L) - \kappa$ , where  $\kappa = \dim(\text{row}(G) \cap U_L)$ . Consequently

$$\mathbb{P}(C_L = 1 \mid A) = 2^{-(\pi(L) - \kappa)}$$

exactly, whenever the exposed cutoff contains the border coordinates. This finite identity needs no rare-intensity regime. If the stack is incompatible, that probability is zero. The parameter  $\kappa$  counts consequences entirely supported on the border coordinates, not input equations that merely mention them.

For the asymptotic conclusions assume  $M, L \rightarrow \infty$ ,  $L \leq A_0 \log M$ ,  $M 2^{-L} \rightarrow 0$ , and eventually  $I \leq V_M - c\nu_M$  for fixed  $c > 0$ . The matrices, their dimensions and right-hand sides may vary with  $M$ . Multiply the quantitative normalized bulk ledger by  $e^I$  before taking limits. The cutoff term is at most  $e^{-c\nu_M + o(\nu_M)}$ ; all polynomial terms still vanish because  $I = O(V_M) = o(\log M)$ . Thus the amplified bulk error is  $o(\lambda)$ ; amplification of an unspecified little-oh alone would not justify this conclusion. Also  $I = O(V_M) = o(\pi(L))$  and  $\kappa \leq r$ . The conditioned microscopic interior mass, relative to  $2^{-\pi(L) + \kappa}$ , is at most

$$\exp(I) 2^{-\delta_0 \pi(L) - \kappa} = o(1).$$

The mesoscopic first term is  $\lambda M^{\delta-1}$  and remains  $o(\lambda)$  after multiplication by  $e^I$ ; its other term is dominated by  $2^{-g_L}$  and remains negligible relative to the border mass. The same one-hit calculation, normalized by  $2^{-\pi(L) + \kappa} + b$ , therefore proves the affine-conditioned crossover with  $\pi(L)$  replaced by  $\pi(L) - \kappa$ . No assumption on future prime coordinates is used for the rare tail, border concentration or location laws.

Finally let  $q_1 < \dots < q_K$  be the first primes after  $L$ , and put  $Q_{L,K} = \text{span}\{e_{q_1}, \dots, e_{q_K}\}$  and  $H_L = \text{row}(G) + U_L$ . If the extended stack is compatible,

$$\mathbb{P}(G_L \geq K \mid A, C_L = 1) = 2^{-[K - \dim(H_L \cap Q_{L,K})]},$$

otherwise this tail probability is zero. Thus the ordinary geometric mark is preserved when, for every fixed  $K$ , the condition  $H_L \cap Q_{L,K} = \{0\}$  holds beyond a threshold that may depend on  $K$ . This is separate from the parameter  $\kappa$  controlling the border mass. Fix  $K$ , use the finite-cylinder representative and joint stable lift, and take the size limit before  $K \rightarrow \infty$ . No common threshold in  $K$  and no neutrality for a growing truncation  $K(M)$  is assumed. The exact rank formula remains the alternative when neutrality fails; no relative estimate for a disappearing mixture component is inferred.

## F Microscopic comparison by largest-odd-prime pivots

We prove Theorem 7.7. Throughout this appendix the hypotheses are (7.15) and (7.17), with  $\Lambda \rightarrow \infty$ . Suppress the index  $M$  on  $I, V, \nu, Y$ , and put  $H = \log M$ ,  $u = u(\nu)$ . Write  $\mathbb{P}_{\mathcal{A}}, \mathbb{E}_{\mathcal{A}}$  for conditioning on  $\mathcal{A}_M$ .

All arithmetic inputs in this proof are established in the present article–companion pair and are identified in the local register below. The final reciprocal bound also follows from the classical largest-prime-factor asymptotics of [10]. We retain a direct uniform Rankin proof using the prime number theorem [18], then spell out the alternative deduction. Neither the reciprocal bound nor the general Palm–Stein principle is claimed as a new analytic tool. Palm–Stein comparison is classical [4, 8]; the finite total-variation inequality is included explicitly.

### F.1 The local signed input register

Put  $E_* = E_M^*$  from (7.19),  $Q = L + E_* + 1$ , and  $U = \{x \in \mathbb{Z} : \lceil \sqrt{M} \rceil \leq x \leq n + 1\}$ . A maximal support at  $x = j + 1$  is  $\{j, \dots, j + Q\}$ , with  $Q + 1$  raw vertices. Its base support has  $L + 1$  vertices; these two lengths are not interchanged. Let  $\mathcal{D}_Y(Q)$  be the indices  $j$  with  $j + 1 \in U$  whose maximal support contains a  $Y$ -defective integer. Put

$$\mathcal{M}_{L+1} = \sum_{x \in U} (2^{m_{L+1}(x)} - 1), \quad R_{2,U}^{\text{val}}(M, Q + 1) = \sum_{\substack{x, y \in U \\ |x-y| > Q}} (2^{\rho_{Q+1}^{\text{val}}(x,y)} - 1).$$

The full-value kernel is the one defined in section 3.7, with no block-parity restriction.

**Lemma F.1** (Signed inputs and an independent shifted scalar tail). *Under (7.15) and (7.17),  $Q = \log_2 M + O(V)$  and  $Y > 2(Q + 1)$  eventually. Uniformly in this band,*

$$\mathcal{M}_{L+1} \leq M^{1/2+o(1)}, \quad |\mathcal{D}_Y(Q)|/M \leq e^{-V+o(\nu)}, \quad (\text{F.1})$$

$$R_{2,U}^{\text{val}}(M, Q + 1) \ll_{\varepsilon} M^{\varepsilon} (M^{5/3} + M^{2/3} 2^{Q+1}). \quad (\text{F.2})$$

*On every retained support the signed low-type marginal is exactly  $p\gamma_{e,\sigma}$ , on every environment in  $\mathcal{F}_Y$ . For two retained sites with  $0 < |j - k| \leq Q$ , the conditional joint mass is at most  $4p^2\gamma_{e,\sigma}\gamma_{f,\tau}$ . For separated sites its unconditional counterpart is at most*

$$p^2\gamma_{e,\sigma}\gamma_{f,\tau} 2^{\rho_{Q+1}^{\text{val}}(j+1,k+1)}. \quad (\text{F.3})$$

*Finally, for  $L' = L + E_* + 1$  and  $\lambda' = M2^{-L'}$ ,*

$$\mathbb{P}(\exists 2 \leq x \leq n + 1 : J_{x,L'} = 1) \ll \lambda' + e^{-c_0 \log M / \log \log M}, \quad e^I \lambda' \leq e^{-V}. \quad (\text{F.4})$$

*This scalar tail is established without the microscopic comparison.*

*Proof.* Write  $\lambda = Mp$  for the ambient intensity in the cutoff, whereas  $\Lambda = np$  is the exact target intensity in the hypothesis. Since  $L = O(\log M)$ , eventually  $\Lambda \leq \lambda \leq 2\Lambda$ . For any fixed  $0 < c_1 < c$ , the cost  $\log 2$  is absorbed by  $(c - c_1)\nu$ , giving  $I + \log \lambda \leq V - c_1\nu$  eventually. Thus no stronger conditioning hypothesis is introduced. The cutoff gives  $2^{E_*} \geq e^{I+V}(2 + \lambda)$  and, in particular,  $e^I \lambda 2^{-E_*-1} \leq e^{-V}/2$ . The budget also gives  $I, \log \Lambda, a_M, E_* = O(V) = o(\log M)$ . Proposition 2.4 and Corollary 2.5, including the macroscopic form (2.4), give the weighted mass and, for a deterministic mask  $A \subset U$ , the bound

$$\sum_{x \in A} \mathbb{P}(J_{x,L} = 1) \leq p|A| + p\mathcal{M}_{L+1}.$$

Propositions B.1 and 4.2 apply to raw heights  $M + O(\log M)$  and give the bad-support count. The factor  $Q + 1$  and changes of this size in the ambient height have logarithm  $o(\nu)$ . Proposition 3.24 restricted to  $U$  gives the displayed full-value envelope. It is specialized to  $M^{5/3+o(1)}$  only after  $Q = \log_2 M + O(V)$  has been checked.

For the signed marginals and local pairs, apply section C.2. A prime above  $Y > 2(Q + 1)$  occurring oddly at one vertex is private throughout a local union. Its raw values are thus independent fair signs conditionally on the small-prime environment. For  $j < k$ ,  $a = L + e$  and  $d = k - j$ , the ratios to  $p^2 \gamma_{e,\sigma} \gamma_{f,\tau}$  are

$$0 \ (d < a), \quad 4\mathbf{1}_{\{\tau=-\sigma\}} \ (d = a), \quad 2\mathbf{1}_{\{\tau=\sigma\}} \ (d = a + 1), \quad 1 \ (a + 1 < d \leq Q).$$

Zero extension to the maximal supports gives (F.3); it is an unconditional, not a fibrewise, profile bound. For a general  $\mathcal{A}_M$ , nonnegative averaged errors may be multiplied by  $e^I$  as in Lemma 6.1. The local bound by four times the rate product already holds on every small-prime fibre and therefore also after mixing under  $\mathcal{A}_M$ , without that factor. The later common factor  $e^I$  is a valid, coarser way of writing the combined ledger.

Proposition 7.3 gives the independent deletion of  $x < \lceil \sqrt{M} \rceil$ , of mass  $O(\Lambda M^{-1/2} + e^{-c_0 \log M / \log \log M})$ . For the shifted tail,  $L'$  remains in a fixed logarithmic band. Apply Theorem 7.4 to its count over  $2 \leq x < M$  and then test non-vacancy. Since  $1 - e^{-\lambda'} \leq \lambda'$ , this gives (F.4); the retained indices are a subset of that range. Its proof uses the scalar graph estimate, the two-window profile and proposition 7.3, not the theorem being proved here. Lastly (7.19) gives  $e^I M p 2^{-E_*-1} \leq e^{-V}$  directly. Summing signs does not enlarge the tail event.  $\square$

## F.2 A finite conditional coupling inequality

Let  $\mathcal{J}$  be finite, let  $W = (W_a)_{a \in \mathcal{J}}$  be indicators with positive means  $\lambda_a$ , and for each  $a$  couple  $W^a$  to  $W$  with  $\mathcal{L}(W^a) = \mathcal{L}(W \mid W_a = 1)$ . Denote a coordinate unit vector by  $e_a$ .

**Lemma F.2** (Dimension-free Palm–Stein bound). *With the supremum convention for total variation,*

$$d_{\text{TV}} \left( \mathcal{L}(W), \bigotimes_a \text{Pois}(\lambda_a) \right) \leq \sum_a \lambda_a \mathbb{E} \|W - (W^a - e_a)\|_1. \quad (\text{F.5})$$

*No dependency graph is assumed.*

*Proof.* The immigration–death generator with invariant product-Poisson law  $\mathbf{Q}$  is

$$\mathcal{A}g(w) = \sum_a \lambda_a \Delta_a g(w) + \sum_a w_a \{g(w - e_a) - g(w)\}.$$

For  $0 \leq h \leq 1$ , its semigroup solution  $g_h(w) = -\int_0^\infty (\mathbb{E}h(Z_t^w) - \mathbf{Q}h) dt$  satisfies  $\mathcal{A}g_h = h - \mathbf{Q}h$ . The integral is finite by coupling the initial particles to a stationary initial configuration; the expected number of unmatched survivors is at most  $(|w| + \sum_a \lambda_a)e^{-t}$ . Couple two extra initial particles independently, each with death rate one. Their simultaneous survival probability is  $e^{-2t}$  and the absolute second difference of an indicator is at most two. Hence

$$|\Delta_a \Delta_b g_h(w)| \leq \int_0^\infty 2e^{-2t} dt = 1, \quad |\Delta_a g_h(w) - \Delta_a g_h(v)| \leq \|w - v\|_1.$$

The first inequality includes  $a = b$ . For the second, put  $u = w \wedge v$ , the coordinatewise minimum. Telescope from  $u$  to  $w$  and from  $u$  to  $v$  along coordinate steps. Both paths stay in the nonnegative lattice and their total length is  $\|w - u\|_1 + \|v - u\|_1 = \|w - v\|_1$ . Thus comparability of  $w$  and

$v$  is not required. The conditional-law identity gives

$$\mathbb{E} \mathcal{A} g_h(W) = \sum_a \lambda_a \mathbb{E} \{ \Delta_a g_h(W) - \Delta_a g_h(W^a - e_a) \}.$$

Here  $W^a - e_a \geq 0$ . Bound the differences and take the supremum over indicator tests. All sums are finite.  $\square$

For categorical indicators  $W_{j,\alpha}$ , suppose forcing  $(j, \alpha)$  makes it the unique occupied type at  $j$  and leaves every site outside a deterministic set  $\mathcal{D}_j$  unchanged. Put  $s_j = \sum_\alpha \lambda_{j,\alpha}$ . The same-site part of (F.5) is  $\sum_j s_j^2$ . The off-site part is at most

$$\sum_j \sum_{k \in \mathcal{D}_j \setminus \{j\}} \left\{ s_j s_k + \sum_{\alpha, \alpha'} \mathbb{P}(W_{j,\alpha} = W_{k,\alpha'} = 1) \right\}. \quad (\text{F.6})$$

Indeed  $|W_{k,\alpha'} - W_{k,\alpha'}^{j,\alpha}| \leq W_{k,\alpha'} + W_{k,\alpha'}^{j,\alpha}$ , and  $\lambda_{j,\alpha} \mathbb{E} W_{k,\alpha'}^{j,\alpha} = \mathbb{P}(W_{j,\alpha} = W_{k,\alpha'} = 1)$ . Thus a very small type probability cancels, rather than appearing as an inverse minimum probability. The analytic solution and its Hessian bound are the explicit input used by the associated Lean proof of Lemma F.2; the semigroup construction printed above is not reconstructed by that formal proof. This distinction concerns formal coverage, not an extra hypothesis in the written coupling argument.

### F.3 A uniform reciprocal-pivot estimate

For  $m \geq 1$  define  $\kappa_{\text{odd}}(m)$  as in the article and put  $C(X, z) = \#\{1 \leq m \leq X : \kappa_{\text{odd}}(m) \leq z\}$ . For  $z \geq 1$ , the condition is exactly that every valuation at a prime above  $z$  is even, namely the earlier defective-integer predicate. The integer zero is never included. The pivot is not the ordinary largest prime factor: for instance  $\kappa_{\text{odd}}(2 \cdot 13^2) = 2$ . Equivalently  $\kappa_{\text{odd}}(m) = P^+(\text{core}(m))$ , with  $P^+(1) = 1$ ; this is  $P^*(m)$  in [6, eq. (5)]. Writing  $m = a^2 v$  with  $v$  squarefree gives, for  $s > 1/2$ ,

$$\sum_{\kappa_{\text{odd}}(m) \leq z} m^{-s} = \zeta(2s) \prod_{q \leq z} (1 + q^{-s}), \quad C(X, z) \leq X^s \zeta(2s) \prod_{q \leq z} (1 + q^{-s}). \quad (\text{F.7})$$

The square part includes primes larger than  $z$  with even exponent and cannot be omitted.

**Lemma F.3** (Uniform Rankin estimate). *For  $X = M + O(\log M)$ , uniformly for  $V - 2 \leq w \leq 3V + 2$ ,*

$$C(X, e^w) \leq X \exp\{-D(H/w) + o(\nu)\}. \quad (\text{F.8})$$

*The remainder is deterministic and uniform in this band.*

*Proof.* Put  $v = u(H/w)$  and  $s = 1 - v/w$ . Throughout the band,  $H/w \asymp \nu$ ,  $v = u + O(1)$ , and  $s \rightarrow 1$ , so eventually  $s \geq 3/4$ . Therefore

$$\log \left\{ \zeta(2s) \prod_{q \leq e^w} (1 + q^{-s}) \right\} = \sum_{q \leq e^w} q^{-s} + O(1).$$

The sum is  $\text{Ei}(v) + o(\nu)$  uniformly. To check the required precision, split at  $e^{w/2}$ . Chebyshev's bound and partial summation bound the lower part by  $O(\log w + e^{v/2}) = o(\nu)$ . On the upper part the relative error in  $\pi(t) \sim t/\log t$  is uniform for  $t \geq e^{(V-2)/2}$  and tends to zero. Partial

summation compares the sum to

$$\int_{e^{w/2}}^{e^w} \frac{t^{-s}}{\log t} dt = \text{Ei}(v) - \text{Ei}(v/2).$$

This integral is  $O(\nu)$ . The boundary terms  $O(e^v/w + e^{v/2}/w)$ , the replacement of a partial-summation factor  $s$  by one, and  $\text{Ei}(v/2)$  are all  $o(\nu)$ . Since  $\log X = H + o(1)$ , the logarithm of the normalized bound in (F.7) is

$$(s-1)\log X + \text{Ei}(v) + o(\nu) = -(H/w)v + \text{Ei}(v) + o(\nu) = -D(H/w) + o(\nu).$$

□

**Lemma F.4** (Convex saddle envelope). *In the preceding band,*

$$w + D(H/w) \geq 2V - V \left( \sqrt{\nu u/V} - 1 \right)^2 = 2V - O(\nu/u) = 2V - o(\nu). \quad (\text{F.9})$$

*Proof.* A parameter calculation gives the tangent inequality directly. Put  $\nu(t) = e^t/t$  and  $D_{\text{par}}(t) = e^t - \text{Ei}(t)$  for  $t > 1$ . At the fixed saddle parameter  $u$ ,

$$\frac{d}{dt} \{D_{\text{par}}(t) - u\nu(t)\} = \frac{e^t(t-1)(t-u)}{t^2}.$$

This expression is nonpositive up to  $u$  and nonnegative afterwards, so the bracket has its minimum at  $u$ . Since  $D_{\text{par}}(u) = V$  and  $\nu(u) = \nu$ , evaluation at the parameter corresponding to  $H/w$  gives  $D(H/w) \geq V + u(H/w - \nu)$ , without differentiating an inverse function. With  $a = \nu u/V$  and  $t = w/V$ , add  $w$  and minimize  $t + a/t$ :

$$w + D(H/w) \geq V(t + 1 - a + a/t) \geq V(1 - a + 2\sqrt{a}) = 2V - V(\sqrt{a} - 1)^2.$$

Now  $\nu u - V = \text{Ei}(u) = \nu(1 + O(1/u))$  and  $V \sim \nu u$ , so

$$V(\sqrt{a} - 1)^2 = \frac{\text{Ei}(u)^2}{(\sqrt{\nu u} + \sqrt{V})^2} = O(\nu/u).$$

The minimizer need not be exactly  $w = V$ ; the displayed precision, not exact optimality of that choice, preserves the fixed margin  $c\nu$ . □

**Proposition F.5** (Reciprocal-pivot sum). *For  $X = M + O(\log M)$  and  $Y = \lfloor e^V \rfloor$ ,*

$$S_M(X) := \sum_{\substack{m \leq X \\ \kappa_{\text{odd}}(m) > Y}} \frac{1}{\kappa_{\text{odd}}(m)} \leq X \exp\{-2V + o(\nu)\}. \quad (\text{F.10})$$

*Proof.* Terms with  $\kappa_{\text{odd}}(m) > e^{3V}$  contribute at most  $Xe^{-3V}$ . Divide the remaining logarithmic range into intervals  $[b, b+1)$ , starting at  $\log Y = V + o(1)$ . Each contributes at most

$$e^{-b}C(X, e^{b+1}) \leq X \exp\{-b - D(H/(b+1)) + o(\nu)\} \leq X \exp\{-2V + o(\nu)\},$$

by Lemmas F.3 and F.4. Indeed  $-b - D(H/(b+1)) = 1 - \{b+1 + D(H/(b+1))\}$ : the next-slice shift costs the fixed factor  $e$  before absorption in  $o(\nu)$ . For an integer pivot,  $\kappa_{\text{odd}} > \lfloor e^V \rfloor$  is exactly  $\kappa_{\text{odd}} > e^V$ ; no threshold mass is lost. There are  $O(V)$  intervals, with  $\log V = o(\nu)$ . Their uniform remainders can be summed. The omitted tail is smaller since  $V/\nu \rightarrow \infty$ . □

**Classical antecedent and square-part reduction.** The bound in proposition F.5 is not an autonomous analytic novelty. With  $P^+(1) = 1$ , write  $S_+(x) = \sum_{n \leq x} 1/P^+(n)$ . Equation (1.3) of [10], at  $r = 1$ , gives

$$S_+(x) = x \exp\{-G(\log x) + o(\nu_x)\}, \quad G(H) = \sqrt{2H(\log H + \log \log H - \log 2 - 2)}.$$

The expansion of  $V_x$  in proposition 4.2 gives  $G(\log x) = 2V_x + o(\nu_x)$ . In the classical formula, the omitted terms after the first correction have size  $O(\sqrt{H \log H}(\log \log H)^2/(\log H)^2) = o(\nu_x)$ . This identifies the required second-scale precision, not just the leading square-root exponent.

Here is the deduction for odd-prime pivots. Uniquely write  $n = a^2 v$  with  $v$  squarefree. Positivity gives

$$\sum_{n \leq x} \frac{1}{\kappa_{\text{odd}}(n)} = \sum_{a \leq \sqrt{x}} \sum_{\substack{v \leq x/a^2 \\ v \text{ squarefree}}} \frac{1}{P^+(v)} \leq \sum_{a \leq \sqrt{x}} S_+(x/a^2).$$

The terms  $a > \exp(3V_x)$  are  $O(xe^{-3V_x})$  by  $S_+(t) \leq t$ . For the other terms, put  $H = \log x$  and  $t = x/a^2$ . Uniformly here,  $\log t = H - O(V_x)$ ,  $\nu_t \sim \nu_x$ , and the remainder in the classical asymptotic is  $o(\nu_x)$ . The explicit smooth function  $G$  satisfies  $G'(H) = O(\sqrt{\log H/H}) = o(1)$  throughout this range. The mean-value theorem therefore yields

$$S_+(x/a^2) \leq x \exp\{-2V_x + o(\nu_x)\} a^{-2+o(1)}.$$

Only the explicit function  $G$  has been differentiated, not an unspecified asymptotic remainder. The weights sum to  $O(1)$ , so

$$\sum_{n \leq x} \frac{1}{\kappa_{\text{odd}}(n)} \leq x \exp\{-2V_x + o(\nu_x)\}.$$

Restricting to  $\kappa_{\text{odd}}(n) > Y$  only decreases this sum. At  $x = M + O(\log M)$ , the hard scales agree at the stated precision, recovering proposition F.5. This is a deduction from the cited asymptotic, not an assertion that its statement uses the present pivot notation. The direct Rankin argument above remains a complete proof of the estimate used below.

#### F.4 Directed change sets and exact forcing

Take  $E_* = E_M^*$  from (7.19),  $Q = L + E_* + 1$ , and  $\mathcal{T}_* = \{0, \dots, E_*\} \times \{-1, +1\}$ . Use the original deterministic good set

$$G_0 = \{j : 1 \leq j \leq n, j+1 \geq \lceil \sqrt{M} \rceil, \kappa_{\text{odd}}(j+a) > Y \ (0 \leq a \leq Q)\}. \quad (\text{F.11})$$

For  $j \in G_0$ , put  $q_{j,a} = \kappa_{\text{odd}}(j+a)$ . These primes are distinct and greater than  $Y > 2(Q+1)$ : a selected prime divides its own vertex to an odd exponent and no other vertex of that maximal support. For this privacy assertion alone, exceeding  $Q$  suffices: a common prime divisor of two distinct vertices would divide their nonzero difference of size at most  $Q$ . Odd valuation, not valuation exactly one, supplies the parity coordinate. The stronger  $Y > 2(Q+1)$  is retained for the separate local-pair argument.

The finite cylinder must include every prime up to an integer  $R \geq \max\{Y, n+Q\}$ . This covers all marked vertices and the entire conditioning event, not only the terminal prefix  $M$ . Restricting the infinite prime-sign source to this cylinder preserves the conditional laws of all

the finite events below. Define

$$\mathcal{D}_j = \{k \in G_0 : q_{j,a} \mid k + b \text{ for some } 0 \leq a, b \leq Q\}, \quad \mathcal{B}_M = \sum_{j \in G_0} |\mathcal{D}_j|. \quad (\text{F.12})$$

These are the possible changes for a specified forcing, not neighbourhoods asserted to be a new dependency graph. Divisibility, including even exponents, is a harmless overestimate of possible change.

**Proposition F.6** (The total directed footprint). *With  $X = n + Q = M + O(\log M)$ ,*

$$|\mathcal{D}_j| \leq (Q + 1) \sum_{a=0}^Q (n/q_{j,a} + 1), \quad (\text{F.13})$$

$$\mathcal{B}_M \leq n(Q + 1)^2(1 + S_M(X)) \ll n^2 e^{-2V+o(\nu)} + n(Q + 1)^2. \quad (\text{F.14})$$

*Proof.* For a fixed prime and target offset  $b$ , at most  $n/q + 1$  grid indices satisfy the divisibility; no bound on  $b/q$  is required. Sum over offsets and selected primes. When summing reciprocals over  $j, a$ , each raw integer  $m = j + a$  occurs at most  $Q + 1$  times. Apply Proposition F.5, use  $X \asymp n$ , and absorb  $\log(Q + 1) = O(\log H) = o(\nu)$ .  $\square$

For example, let  $Q = 3$ ,  $Y = 11$ ,  $j = 246$  and  $k = 101$ . The source vertices 246, 247, 248, 249 have pivots 41, 19, 31, 83. The all-large-prime graph connects them to 101, 102, 103, 104 through 13, since  $247 = 13 \cdot 19$  and  $104 = 8 \cdot 13$ . None of the four source pivots divides a target vertex, so  $k \notin \mathcal{D}_j$ . Conversely the pivot 13 of 104 divides 247; the direction matters.

**Proposition F.7** (Exact one-word conditional law). *Let  $W = (I_{j,\alpha})_{j \in G_0, \alpha \in \mathcal{T}_*}$ . For every  $a = (j, e, \sigma)$ , there is a deterministic modification of prime signs under  $\mathbb{P}_{\mathcal{A}}$  whose resulting vector  $W^a$  has law*

$$\mathcal{L}_{\mathcal{A}}(W^a) = \mathcal{L}_{\mathcal{A}}(W \mid I_{j,e,\sigma} = 1).$$

*Only  $q_{j,0}, \dots, q_{j,L+e+1}$  are modified. The small-prime environment and every coordinate at a site outside  $\mathcal{D}_j$  remain unchanged. If the word already holds, the modification is the identity.*

*Proof.* Write  $f(q) = (-1)^{\eta_q}$  and  $\sigma = (-1)^\tau$ . The required raw word has bits  $b_0 = b_{L+e+1} = 1 - \tau$  and  $b_a = \tau$  for  $1 \leq a \leq L + e$ . By privacy, the raw bit at  $j + a$  is  $\eta_{q_{j,a}} + \ell_{j,a}(\eta_{\text{nonpivots}})$ , where  $\ell_{j,a}$  uses none of the pivot bits being forced. Keep the other bits fixed and set

$$\eta'_{q_{j,a}} = b_a + \ell_{j,a}(\eta_{\text{nonpivots}}) \pmod{2}, \quad 0 \leq a \leq L + e + 1. \quad (\text{F.15})$$

Every other bit is left unchanged, including the pivots with  $L + e + 2 \leq a \leq Q$  in the maximal support. Conditional on all the nonpivot bits and on  $\mathcal{A}_M$ , the distinct pivot bits are independent and fair. Thus the event has probability  $2^{-(L+e+2)} = p\gamma_{e,\sigma}$  for every nonpivot configuration. Conditioning on it leaves the complete nonpivot law unchanged and fixes exactly the values in (F.15). This proves the joint conditional law of the entire finite vector. All forced primes exceed  $Y$ . Outside  $\mathcal{D}_j$ , none divides any of the raw vertices of a low-type event, so that event is unchanged. Categorical exclusivity gives the unique forced atom at the source site.  $\square$

This is the step that deals with higher-order dependence: a complete conditional law is constructed, not inferred from pair covariances. The estimates below use pairs only after that law and its possible changes have been established.

### F.5 Finite signed ledger and completion

At each  $j \in G_0$ , every low-type conditional mean is  $p\gamma_\alpha$  on every small-prime environment and hence under  $\mathbb{P}_A$ . Apply Lemma F.2 and (F.6) to the preceding forcing. The same-site sum is at most  $np^2$  and the product-of-means sum is at most  $p^2\mathcal{B}_M$ . For  $0 < |j - k| \leq Q$ , the signed local calculation in Lemma F.1 bounds the joint mass by  $4p^2\gamma_\alpha\gamma_{\alpha'}$ , already uniformly over environments. There are  $O(nQ)$  such pairs. For separated sites the unconditional bound is

$$\mathbb{P}(I_{j,\alpha} = I_{k,\alpha'} = 1) \leq p^2\gamma_\alpha\gamma_{\alpha'}2^{\rho_{Q+1}^{\text{val}}(j+1,k+1)}.$$

Bound its conditional version by  $e^I$  times this nonnegative quantity, split  $2^\rho = 1 + (2^\rho - 1)$ , and enlarge only the excess to all separated pairs. The constant term stays on the directed change sets; extending it to all pairs would add an unnecessary quadratic baseline. Since  $\sum_{\alpha \in \mathcal{T}_*} \gamma_\alpha \leq 1$ , this proves

$$d_{\text{TV}} \left( \mathcal{L}_A(W), \bigotimes_{j \in G_0, \alpha \in \mathcal{T}_*} \text{Pois}(p\gamma_\alpha) \right) \ll np^2 + e^I p^2 \{nQ + \mathcal{B}_M + R_{2,U}^{\text{val}}(M, Q + 1)\}. \quad (\text{F.16})$$

Neither the number of exact types nor their smallest probability appears.

Embed the finite field by zero elsewhere. The total actual and target deletion-and-tail cost is bounded by a fixed multiple of

$$d_{0,M} = e^I \left[ p\mathcal{M}_{L+1} + p|\mathcal{D}_Y(Q)| + \Lambda M^{-1/2} + e^{-c_0 H / \log H} + Mp 2^{-E_* - 1} \right]. \quad (\text{F.17})$$

This follows from the masked first moment, independent deep-start bound and shifted scalar tail in Lemma F.1. The actual upper-tail event is a base hit at  $L + E_* + 1$ ; signs do not enlarge it. The independent target cost is its omitted intensity. Filling is unnecessary for (F.16), because the Stein bound is dimension free for the reduced deterministic target. Thus

$$\Delta_M^{\text{mic}} \ll d_{0,M} + np^2 + e^I p^2 \{nQ + \mathcal{B}_M + R_{2,U}^{\text{val}}(M, Q + 1)\}. \quad (\text{F.18})$$

By (F.14), the footprint term is at most

$$e^{I+2\log \Lambda - 2V + o(\nu)} + M^{-1+o(1)} \leq e^{-I-2c\nu+o(\nu)} + M^{-1+o(1)}.$$

The deletion term is bounded by

$$e^{I+\log \Lambda - V + o(\nu)} + M^{-1/2+o(1)} + e^{I-c_0 H / \log H} + O(e^{-V}).$$

Here  $H / \log H \gg V$  makes the deep-start term negligible. Since  $I, \log \Lambda, E_* = O(V) = o(H)$  and  $Q = \log_2 M + O(V)$ , the full-value input gives

$$e^I p^2 R_{2,U}^{\text{val}}(M, Q + 1) \ll_{\varepsilon_0} M^{-1/3+\varepsilon_0+o(1)}.$$

The local and same-site terms are  $M^{-1+o(1)}$ . Choose  $\varepsilon_0 < \varepsilon$  and absorb the uniform  $o(\nu)$  terms with  $c' < c$ . This proves (7.18).

**Dependency boundary.** This proof uses a single forced word at a time. It does not condition on a Poisson-size plant, estimate an exponentially small planted void, or bound a global absolute cumulant activity. The constructions in Appendix G are separate consequences and comparisons. The selected-pivot footprint replaces the all-large-prime baseline  $e^{I+2\log \Lambda - V + o(\nu)}$ ; there is no additional inverse-intensity Stein factor.

## G Private kernels, signed Palm voids, and the absolute-cumulant obstruction

This appendix is a methodological complement for the same binary model. It retains three distinct conclusions: regular configurations drawn from the target, finite signed Palm identities, and a counterexample to a global absolute-cumulant smallness condition. All arguments are given below. The one-sided Palm deficit follows *from* Theorem 7.7, never the reverse. None of these constructions is an input to section F. Keep the hard scales, cutoff  $E_*$ , maximal support diameter  $Q$ , original good set  $G_0$  and conditional law  $\mathbb{P}_A$  used there, and put  $\beta = \log 2$ .

### G.1 Odd rough kernels and a stronger good set

Define

$$r_Y(m) = \prod_{\substack{q > Y \\ v_q(m) \text{ odd}}} q, \quad Z_Y(s) = \zeta(2s) \prod_{q \leq Y} (1 + q^{-s}), \quad B(X, Y; s) = X^{s-1} Z_Y(s), \quad \frac{1}{2} < s < 1.$$

The product  $r_Y(m)$  and the single pivot  $\kappa_{\text{odd}}(m)$  have different roles. Every integer has a unique form  $m = a^2 v r$ , with  $v$  squarefree and  $Y$ -smooth and  $r = r_Y(m)$  squarefree with primes above  $Y$ . If  $D_Y(x) = \#\{m \leq x : r_Y(m) = 1\}$ , Rankin gives  $D_Y(x) \leq x^s Z_Y(s)$ . The exact convolutions over squarefree rough  $r$  imply

$$\frac{1}{X} \#\{m \leq X : r_Y(m) \leq T\} \leq B(X, Y; s) \left(1 + \frac{T^{1-s}}{1-s}\right), \quad (\text{G.1})$$

$$\begin{aligned} \frac{1}{X} \sum_{\substack{m \leq X \\ r_Y(m) > 1}} \frac{z^{\omega(r_Y(m))}}{r_Y(m)} &\leq B(X, Y; s) \left\{ \prod_{q > Y} (1 + z q^{-1-s}) - 1 \right\} \\ &\leq B(X, Y; s) \{\exp(z/(sY^s)) - 1\}, \quad z \geq 1. \end{aligned} \quad (\text{G.2})$$

For the first inequality sum  $D_Y(X/r)$  over rough  $r \leq T$  and enlarge to all positive integers. For the second sum  $z^{\omega(r)} D_Y(X/r)/r$ ; the empty kernel is removed before forming the Euler product. Finally  $\sum_{q > Y} q^{-1-s} \leq \sum_{m > Y} m^{-1-s} \leq Y^{-s}/s$ .

At  $s = 1 - u/V$  and  $X = M + O(Q)$ , the calculation of Lemma F.3 at  $w = V$  gives

$$B(X, Y; s) \leq e^{-V+o(\nu)}, \quad Y^{-s} = e^{-V+u+o(1)}. \quad (\text{G.3})$$

Fix  $0 < \vartheta < c$ , and put

$$T_M = \exp(\vartheta H/u), \quad G_\vartheta = \{j \in G_0 : r_Y(j+a) > T_M \ (0 \leq a \leq Q)\}. \quad (\text{G.4})$$

This is a threshold on a product of primes, not a change of the conditioning cutoff  $Y$ . Eventually  $T_M > Y$  and  $T_M = M^{o(1)}$ .

**Proposition G.1** (Cost of stronger deletion). *One has*

$$p|G_0 \setminus G_\vartheta| \leq \exp\{\log \Lambda - V + \vartheta \nu + o(\nu)\} = o(1), \quad (\text{G.5})$$

and replacing these extra deleted low-type coordinates by independent target coordinates costs at most  $2p|G_0 \setminus G_\vartheta|$  under  $\mathbb{P}_A$ . Moreover

$$|G_\vartheta^c| \leq O(\sqrt{M}) + M e^{-V+\vartheta \nu + o(\nu)} = o(M/H). \quad (\text{G.6})$$

Completing the original deletions and mark tails has cost  $O(d_{0,M} + p|G_0 \setminus G_\vartheta|) = o(1)$ .

*Proof.* Each integer is used in at most  $Q + 1$  maximal supports. Use (G.1)–(G.3) and  $(1 - s) \log T_M = \vartheta \nu$ ; the logarithms of  $Q + 1$  and  $1/(1 - s)$  are  $O(\log H) = o(\nu)$ . This proves the count and the first bound, whose exponent is at most  $-I - (c - \vartheta)\nu + o(\nu)$ . At a site of  $G_0$ , each low-type mean is exactly  $p\gamma_\alpha$  on every environment. Thus the added real and target masses are each at most  $p$  per site without an extra factor  $e^I$ . The original deletion and high marks are covered by (F.17). The last equality in (G.6) follows from  $V/\nu \sim u \rightarrow \infty$  and  $\log H = o(\nu)$ .  $\square$

Call a collection of maximal supports *private-pivot regular* if each raw vertex occurrence has a prime above  $Y$ , occurring there to an odd exponent, that divides no other vertex occurrence of the collection. Repeated vertices preclude regularity.

**Lemma G.2** (CRT allocation of all odd prime factors). *Assume  $n \geq 1$ ,  $Q \leq n$  and  $Y > Q$ , as holds eventually in the ambient regime. Let  $J_1, \dots, J_k$  be independent uniform indices in  $\{1, \dots, n\}$ . Conditional on  $J_i = j$  and on a source vertex  $m = j + a$  with  $r = r_Y(m) > 1$ , the probability that every prime of  $r$  is hosted in one of the other  $k - 1$  supports is at most*

$$\frac{\{3(k - 1)(Q + 1)\}^{\omega(r)}}{r}. \quad (\text{G.7})$$

For  $k \geq 2$  and  $z = 3k(Q + 1) < Y$ , the probability that all indices lie in  $G_\vartheta$  but their supports are not regular is at most  $k(Q + 1)T_M^{-1 + \log z / \log Y}$ . A single good support is regular.

*Proof.* Assign each prime of  $r$  to a target block and offset where it is hosted. There are at most  $((k - 1)(Q + 1))^{\omega(r)}$  assignments. For a fixed assignment, CRT fixes each target index modulo the product  $r_h$  of its assigned primes. A target block receiving no prime has probability exactly one. For a nonempty block,  $r_h \leq r \leq m \leq n + Q \leq 2n$ , so that residue has probability at most  $1/r_h + 1/n \leq 3/r_h$ . There are at most  $\omega(r)$  nonempty blocks; their nontrivial moduli have product  $r$ . Independence therefore bounds the assignment probability by  $3^{\omega(r)}/r$ , not by a factor three for every empty block as well. Summing assignments gives (G.7). If  $r > T_M$ , then  $\omega(r) \leq \log r / \log Y$  and  $z^{\omega(r)}/r \leq T_M^{-1 + \log z / \log Y}$ . Union-bound over the  $k(Q + 1)$  source occurrences. A prime above  $Y > Q$  cannot be hosted elsewhere inside its own support.  $\square$

Let  $\mathcal{R}_M$  be the target configurations of mass at most  $K_M = \lceil 2\Lambda \rceil$ , with low types  $e \leq E_*$ , sites in  $G_\vartheta$ , and regular maximal supports. Equivalently, enumerate all its atoms, with their multiplicities, and impose regularity on every raw vertex occurrence of that enumeration. This definition is invariant under permutation, so it passes from the ordered sample to the point configuration without losing repeated occurrences. The empty configuration is included.

**Theorem G.3** (Private pivots for a Poisson-size target cloud). *For the full same-grid target law  $\mathbf{Q}_M$ ,*

$$\begin{aligned} \mathbf{Q}_M(\mathcal{R}_M^c) &\leq \varepsilon_{\text{reg},M} \ll \Lambda M^{-1/2} + e^{\log \Lambda - V + \vartheta \nu + o(\nu)} + \Lambda 2^{-E_* - 1} \\ &\quad + e^{-(2 \log 2 - 1)\Lambda} + e^{-c\vartheta \nu^2 / (4u)} = o(1). \end{aligned} \quad (\text{G.8})$$

For every regular  $z = \{(j_i, \alpha_i) : 1 \leq i \leq k\}$ ,

$$\mathbb{P}(\text{Occ}(z) \mid \mathcal{F}_Y) = \prod_{i=1}^k p\gamma_{\alpha_i} \quad (\text{G.9})$$

on every small-prime environment, and therefore under  $\mathbb{P}_A$ .

*Proof.* Conditional on target mass  $N = k$ , sites are independent uniform grid indices and types have law  $\gamma$ . Exponential Markov bounds give  $\mathbb{P}(N > K_M) \leq e^{-(2 \log 2 - 1)\Lambda}$ . The bad-site and high-mark expected counts give the first three terms of (G.8). For  $k \leq K_M$ , eventually

$$1 - \frac{\log(3K_M(Q+1))}{\log Y} \geq \frac{c\nu}{2V}.$$

Apply Lemma G.2, obtaining a remaining bound  $K_M(Q+1)e^{-c\nu\nu^2/(2u)}$ . Since  $\nu^2/(uV) \rightarrow \infty$  and  $\log(K_M(Q+1)) \leq V + O(\log H)$ , this is at most the final term of (G.8). For a regular configuration choose one private odd prime per raw vertex. The selected columns of the valuation matrix form an identity minor over  $\mathbb{F}_2$  even conditional on all unchosen signs. The raw values are therefore independent and fair. Each exact signed word fixes  $L + e_i + 2$  of them; integrating unused maximal-support vertices proves (G.9).  $\square$

The first assertion is about configurations drawn from the target. The second prescribes presence, not absence of further occurrences. For instance, if  $X_{2i-1} = X_{2i} = B_i$  with independent Bernoulli- $p$  variables, then target configurations using at most one site per pair have asymptotic probability one when  $np^2 \rightarrow 0$  and their prescribed probabilities equal  $p^k$ . Yet when  $np \rightarrow \infty$ , the actual field lies in that regular set only when empty, of probability  $(1-p)^{n/2} \rightarrow 0$ . Its microscopic distance to the Poisson target tends to one. This surrogate is not the arithmetic model.

## G.2 Exact configuration masses and ordinary Palm deletion

Put  $G = G_\vartheta$ ,  $g = |G|$ ,  $\mu = gp$ , and restrict the full arithmetic and target fields to  $G$ , without truncating exact lengths or filling. Denote their laws by  $\mathbf{P}_M^G, \mathbf{Q}_M^G$ , and their distance by  $\Delta_M^G$ . Let  $d_A = \mathbb{P}_{\mathcal{A}}(\text{a base start occurs outside } G)$  and  $\delta_G = p(n-g)$ ; both tend to zero by Proposition G.1 and the original tail estimates. Projection and the deletion coupling give

$$0 \leq \Delta_M^{\text{mic}} - \Delta_M^G \leq d_A + \delta_G. \quad (\text{G.10})$$

Let  $\mathcal{R}_M^G$  be the same regularity conditions inside  $G$ , with mass bounded by  $K_M$ . Its target exceptional probability  $\varepsilon_{\text{reg},G} = o(1)$  follows by restriction from Theorem G.3. For regular  $z$  put  $\lambda_z = \prod_{(j,\alpha) \in z} p\gamma_\alpha$  and

$$v(z) = \mathbb{P}_{\mathcal{A}}(\mathbf{I}_M = z \mid \text{Occ}(z)), \quad v_G(z) = \mathbb{P}_{\mathcal{A}}(\mathbf{I}_M|_G = z \mid \text{Occ}(z)).$$

The regular configurations have distinct sites, so  $\mathbf{Q}_M^G(\{z\}) = e^{-\mu}\lambda_z$  and  $\mathbf{P}_M^G(\{z\}) = \lambda_z v_G(z)$ . The normalized retained void is

$$R_z(t) = \frac{\mathbb{E}_{\mathcal{A}}\left[\prod_{j \in G \setminus \text{supp } z} (1 - tJ_{j+1,L}) \mid \text{Occ}(z)\right]}{(1 - tp)^{g-|z|}}, \quad 0 \leq t \leq 1. \quad (\text{G.11})$$

It is nonnegative and finite;  $p < 1$  and the plant is regular.

**Proposition G.4** (Averaged deletion and the normalized void deficit). *One has*

$$\begin{aligned} & \mathbb{E}_{\mathbf{Q}_M^G} \mathbf{1}_{\mathcal{R}_M^G}(z) e^\mu (v_G(z) - v(z)) \\ &= \mathbb{P}_{\mathcal{A}}(\mathbf{I}_M|_G \in \mathcal{R}_M^G, \text{a start occurs outside } G) \leq d_A. \end{aligned} \quad (\text{G.12})$$

For the normalized void  $R_z$  in (G.11),

$$\left| \Delta_M^{\text{mic}} - \mathbb{E}_{\mathbf{Q}_M^G} \mathbf{1}_{\mathcal{R}_M^G}(z)(1 - R_z(1))_+ \right| \leq d_A + \delta_G + \varepsilon_{\text{reg},G} + O(\Lambda^2/n). \quad (\text{G.13})$$

*Proof.* Presence factorization implies  $\lambda_z(v_G - v) = \mathbb{P}_{\mathcal{A}}(\mathbf{I}_M|_G = z, \mathbf{I}_M \neq z)$ . Multiplication by the target mass and summation over disjoint events proves (G.12). For any countable laws,  $d_{\text{TV}}(\mathbf{P}, \mathbf{Q}) = \sum_z (\mathbf{Q}(z) - \mathbf{P}(z))_+$ . Thus, with  $D_G = \mathbb{E}_{\mathbf{Q}_M^G} \mathbf{1}_{\mathcal{R}_M^G}(1 - e^\mu v_G)_+$ ,

$$0 \leq \Delta_M^{\text{mic}} - D_G \leq d_A + \delta_G + \varepsilon_{\text{reg},G}.$$

Once the exact planted words are fixed, no further retained atom is, almost surely, the event that every remaining base-start indicator is zero. Simultaneous finiteness of the runs gives this equivalence, and both positive-probability conditionings preserve that full-measure event. Hence  $v_G(z) = (1 - p)^{g-k} R_z(1)$ , where  $k = |z| \leq K_M$ . Uniformly on regular configurations,

$$|\mu + (g - k) \log(1 - p)| \leq kp + \frac{gp^2}{2(1 - p)} = O(\Lambda^2/n).$$

For  $b > 0, x \geq 0$ ,  $|(1 - bx)_+ - (1 - x)_+| \leq 1 - e^{-|\log b|} \leq |\log b|$ ; no bound on  $x = R_z(1)$  is needed. This proves (G.13).  $\square$

The argument pays  $d_A$ , not  $e^\Lambda d_A$ . For the full target, the same mass identity gives directly

$$0 \leq \Delta_M^{\text{mic}} - \mathbb{E}_{\mathbf{Q}_M} \mathbf{1}_{\mathcal{R}_M}(z)(1 - e^\Lambda v(z))_+ \leq \varepsilon_{\text{reg},M}.$$

Consequently Theorem 7.7 makes both one-sided deficits vanish. In the full-target form, setting  $L_z = (-\log v(z) - \Lambda)_+$  on regular configurations when  $v(z) > 0$ ,  $L_z = +\infty$  there when  $v(z) = 0$ , and  $L_z = 0$  otherwise, gives  $(1 - e^\Lambda v(z))_+ = 1 - e^{-L_z}$  on the regular set, with  $e^{-\infty} = 0$ . The extended convention is essential at a zero void. Its expectation vanishes exactly when  $L_z \rightarrow 0$  in target probability. This is not uniform control of every Palm void, and it does not result from dividing an absolute approximation by  $e^{-\Lambda}$ .

**Corollary G.5** (Completion of the one-sided Palm deficit). *Under Theorem 7.7,*

$$\mathbb{E}_{\mathbf{Q}_M^G} \left[ \mathbf{1}_{\mathcal{R}_M^G}(z)(1 - R_z(1))_+ \right] \longrightarrow 0. \quad (\text{G.14})$$

*Equivalently the lower relative-void defect tends to zero in target-configuration probability. This is an averaged, one-sided statement, not a uniform relative estimate over every plant.*

*Proof.* Proposition G.4 identifies the left side with  $\Delta_M^{\text{mic}} + o(1)$ , using exact configuration masses and ordinary deletion costs. Apply Theorem 7.7, whose proof does not use that identity.  $\square$

### G.3 The finite signed polynomial, including the environment

Fix a regular plant  $z$  and write  $\mathcal{B}_z = G \setminus \text{supp } z$ ,  $m = g - |z|$ . Let  $\epsilon$  be the bits of primes at most  $Y$ , and  $\xi$  those above  $Y$  among the finitely many queried integers. Successive relative signs give matrices

$$J_{j+1,L} = \mathbf{1}_{\{A_j \epsilon + B_j \xi = b\}}, \quad b = (1, 0, \dots, 0)^\top \in \mathbb{F}_2^L, \quad A_z \epsilon + B_z \xi = b_z$$

for the base event and the planted raw equations, respectively. Regularity makes  $B_z$  full row rank  $q_z = \sum_{(j,e,\sigma) \in z} (L + e + 2)$ . Planting has conditional probability  $2^{-q_z}$  on every environment,

and so leaves the law of  $\epsilon \mid \mathcal{A}_M$  unchanged. Given that environment,  $\xi$  is uniform on the displayed affine solution space. Put  $\hat{\mu}_A(s) = \mathbb{E}[(-1)^{s \cdot \epsilon} \mid \mathcal{A}_M]$ . For  $S \subseteq \mathcal{B}_z$ ,  $S \neq \emptyset$ , define

$$C_z(S) = \sum_{\substack{u_j \in \mathbb{F}_2^L \setminus \{0\}, j \in S \\ v \in \mathbb{F}_2^{d_z}, \sum_j u_j B_j = v B_z}} (-1)^{\sum_j u_j \cdot b + v \cdot b_z} \hat{\mu}_A \left( \sum_j u_j A_j + v A_z \right), \quad C_z(\emptyset) = 1. \quad (\text{G.15})$$

For a given collection  $(u_j)$  there is at most one  $v$ , by row independence of  $B_z$ .

**Proposition G.6** (Signed avoidance and a target-sign bound). *Writing  $\mathbb{E}_z$  for the planted conditional expectation, for  $0 \leq t \leq 1$ ,*

$$\mathbb{E}_z \prod_{j \in S} (J_{j+1,L} - p) = p^{|S|} C_z(S), \quad (\text{G.16})$$

$$R_z(t) = 1 + \sum_{\emptyset \neq S \subseteq \mathcal{B}_z} \left( -\frac{tp}{1-tp} \right)^{|S|} C_z(S). \quad (\text{G.17})$$

These finite identities include singleton corrections and all overlaps. For fixed unsigned regular geometry  $\zeta$ , expand the sign dependence as

$$R_{\zeta, \sigma}(1) = \sum_{D \subseteq [k]} a_D(\zeta) \prod_{i \in D} \sigma_i, \quad \mathcal{V}(\zeta) = \sum_{D \neq \emptyset} a_D(\zeta)^2.$$

Under the independent fair target signs,

$$\mathbb{E}_{\sigma}(1 - R_{\zeta, \sigma}(1))_+ \leq \min \left\{ 1, (1 - a_{\emptyset}(\zeta))_+ + \frac{1}{2} \sqrt{\mathcal{V}(\zeta)} \right\}. \quad (\text{G.18})$$

*Proof.* The character expansion is  $J_{j+1,L} - p = p \sum_{u_j \neq 0} (-1)^{u_j \cdot b} (-1)^{(u_j A_j) \cdot \epsilon + (u_j B_j) \cdot \xi}$ . On the planted affine space a character  $w \cdot \xi$  has zero mean unless  $w = v B_z$ , in which case its mean is  $(-1)^{v \cdot b_z + (v A_z) \cdot \epsilon}$ . Multiplication and environment averaging prove (G.16). Expand  $\prod_j (1 - t J_{j+1,L}) / (1 - tp)^m = \prod_j \{1 - t(J_{j+1,L} - p) / (1 - tp)\}$  to obtain (G.17).

In planted block  $i$ ,  $b_{z,i} = c_i + \tau_i \mathbf{1}$  with  $\sigma_i = (-1)^{\tau_i}$ , where  $c_i$  is one at the two borders and zero inside. Thus grouping terms by the block parities of  $v$  gives the Walsh expansion. Orthogonality gives mean  $a_{\emptyset}$  and variance  $\mathcal{V}$ . For any integrable  $R$  of mean  $a$ ,

$$\mathbb{E}(1 - R)_+ \leq (1 - a)_+ + \mathbb{E}(a - R)_+ = (1 - a)_+ + \frac{1}{2} \mathbb{E}|R - a| \leq (1 - a)_+ + \frac{1}{2} \sqrt{\text{Var } R}.$$

Here  $R \geq 0$ , so the left side is also at most one. This proves (G.18).  $\square$

For the unconditioned model  $\hat{\mu}_A(s) = \mathbf{1}_{\{s=0\}}$ ; for a general admissible event it is not that indicator. Nor are the planted marginal means assumed to equal  $p$ . The coefficients and their signs must be retained before any norm is taken. Theorem 7.7 controls the deficit in (G.13), but does not establish that  $(1 - a_{\emptyset})_+$  or  $\mathcal{V}$  vanish separately. No such stronger assertion is needed.

Two other exact formulations are useful for separating sufficient conditions from the concluded deficit. Cumulants below are defined by finite moment-partition identities. Deterministic recentering leaves orders at least two unchanged, but  $\text{cum}_z(J_{j+1,L} - p) = \mathbb{E}_z J_{j+1,L} - p$  at order one. Since  $p$  is a reference rate, not necessarily the planted mean, these singleton terms must be kept. If  $W_j = J_{j+1,L} - p$  and

$$A_z(t) = \sum_{\emptyset \neq S \subseteq \mathcal{B}_z} \left( \frac{t}{1-tp} \right)^{|S|} |\text{cum}_z(W_j : j \in S)|,$$

then the finite moment-partition expansion gives  $|R_z(t) - 1| \leq e^{A_z(t)} - 1$ , including singleton Palm corrections. Its per-site weight at  $t = 1$  is  $(1 - p)^{-1}$ , not the global weight 2 discussed below. Also, with  $N_z = \sum_{j \in \mathcal{B}_z} J_{j+1,L}$  and the planted law tilted by  $(1 - t)^{N_z} / \mathbb{E}_z(1 - t)^{N_z}$  for  $t < 1$ ,

$$-\frac{d}{dt} \log R_z(t) = \frac{\mathbb{E}_{z,t} N_z}{1 - t} - \frac{mp}{1 - tp}.$$

Integration to one is an improper identity, with value  $+\infty$  when the void is zero. We assert no smallness estimate for either right-hand side.

#### G.4 The absolute conditional pair layer

There is a useful pair estimate independent of the microscopic proof. For two good disjoint prescribed raw words of total length  $q_1 + q_2 = q$ , write  $A\epsilon + B\xi = b$ ,  $r_0 = 2^{-q}$ ,  $\rho_{>} = q - \text{rank } B$  and  $\rho_{\text{all}} = q - \text{rank}[A \ B]$ . If the full system is compatible, a fraction  $a = 2^{\rho_{\text{all}} - \rho_{>}}$  of environments is admissible, and on those environments its conditional probability is  $r_0 2^{\rho_{>}}$ , otherwise zero. Thus

$$\mathbb{E}_\epsilon |\mathbb{P}(B\xi = b - A\epsilon \mid \epsilon) - r_0| = r_0 \{2^{\rho_{\text{all}}} - 1 + 2(1 - 2^{\rho_{\text{all}} - \rho_{>}})\}. \quad (\text{G.19})$$

If the full system is incompatible the left side is  $r_0$ , and in both cases it is at most  $r_0 \{(2^{\rho_{\text{all}}} - 1) + 2\mathbf{1}_{\{\rho_{>} > \rho_{\text{all}}\}}\}$ . In the incompatible case  $\rho_{\text{all}} \geq 1$ .

For a concrete distinction, take  $L = 3$ ,  $e = f = 0$ , both signs positive,  $Y = 11$ , and starts  $x = 92, y = 114$ . The ten raw vertices are  $91, \dots, 95$  and  $113, \dots, 117$ . Each individual rough row rank is five, their combined rough rank is seven, and their full rank is ten. Thus the unconditioned joint mass is  $2^{-10}$ , but its conditional value is  $2^{-7}$  on one eighth of environments and zero elsewhere. The mean absolute conditional covariance is  $7/4096$ .

Let  $H_2^>(M, Q; Y)$  count ordered separated good maximal-support pairs with nonzero rough nullity. Such a relation uses a vertex  $m$  in one support, and every prime of  $r_Y(m)$  must be hosted by the other support. Assign those primes to target offsets and use CRT. With  $X = M + Q + 1$  this gives

$$H_2^>(M, Q; Y) \ll (Q + 1)X \sum_{\substack{m \leq X \\ r_Y(m) > 1}} \frac{(Q + 1)^{\omega(r_Y(m))}}{r_Y(m)}.$$

There is only one target block, whose residue count is  $O(X/r_Y(m))$  since  $r_Y(m) \leq X$ . Equations (G.2)–(G.3) at  $z = Q + 1$  imply

$$H_2^>(M, Q; Y) \leq M^2 e^{-2V + o(\nu)}. \quad (\text{G.20})$$

Indeed  $(Q + 1)Y^{-s} \rightarrow 0$  and  $u + \log Q = o(\nu)$ .

**Proposition G.7** (The full signed pair activity). *Uniformly for deterministic  $G \subseteq G_0$ , the low-type pair activity obeys*

$$\begin{aligned} \mathcal{K}_{2,M} &:= 2 \sum_{\substack{j < k \\ j, k \in G}} \sum_{\alpha, \alpha' \in \mathcal{T}_*} |\text{Cov}_{\mathcal{A}}(I_{j,\alpha}, I_{k,\alpha'})| \\ &\ll e^I p^2 \{nQ + R_{2,U}^{\text{val}}(M, Q + 1) + H_2^>(M, Q; Y)\} \\ &\ll_\epsilon e^{I + 2 \log \Lambda - 2V + o(\nu)} + M^{-1/3 + \epsilon} = o(1). \end{aligned} \quad (\text{G.21})$$

*Proof.* Good-word marginal means are constant on environments. Hence absolute covariance under  $\mathcal{A}_M$  is at most  $e^I$  times mean absolute covariance over the unconditioned small-prime environments. Apply (G.19). Zero extension bounds the full-kernel excess by the maximal

full-value profile; the second term requires a rough relation host. Local pairs cost  $O(e^I p^2 n Q)$  with signed factor at most four. Sum  $\gamma_\alpha \gamma_{\alpha'}$  before applying (G.20) and the profile. The budget bounds the first exponential by  $e^{-I-2c\nu+o(\nu)}$ .  $\square$

This is an absolute conditional covariance result, not a graph theorem: zero pair covariance does not imply independence from families of other sites. In particular it cannot replace the exact one-word coupling in Appendix F.

### G.5 The finite cumulant bound and its arithmetic obstruction

Let  $G \subseteq G_0$  be a deterministic retained set and let  $\mathcal{T}_* = \{0, \dots, E_M^*\} \times \{-1, +1\}$ . Define the global distinct-site activity

$$\mathcal{K}_{\text{abs}, M} = \sum_{\substack{S \subseteq G \\ |S| \geq 2}} 2^{|S|-1} \sum_{\alpha \in \mathcal{T}_*^S} \left| \text{cum}(I_{j, \alpha_j} : j \in S) \right|. \quad (\text{G.22})$$

Here the cumulants are under the stated arithmetic law, not under a product target.

For a finite family of categories  $Y_j \in \{\partial\} \cup \mathcal{T}_j$  with marginals  $\nu_j$ , let  $I_{j,a} = \mathbf{1}_{\{Y_j=a\}}$  and let  $\mathcal{K}$  be the activity (G.22) with these types and their actual law.

**Lemma G.8** (A finite categorical cumulant bound). *One has*

$$d_{\text{TV}} \left( \mathcal{L}((Y_j)_j), \bigotimes_j \nu_j \right) \leq \min\{1, (e^{2\mathcal{K}} - 1)/2\}. \quad (\text{G.23})$$

For  $t \geq 0$ , if  $B_j$  are Bernoulli variables of means  $r_j$ , and  $F_t = \mathbb{E} \prod_j \{1 + t(B_j - r_j)\}$ , then

$$|F_t| \leq \exp \left\{ \sum_{\substack{S \\ |S| \geq 2}} t^{|S|} |\text{cum}(B_j : j \in S)| \right\}. \quad (\text{G.24})$$

For categorical occupancies  $B_j = \sum_a I_{j,a}$  with  $2 \max_j r_j < 1$ ,  $\mathcal{K} \geq \frac{1}{2} \log F_2$ .

*Proof.* The exact identity  $\delta_{Y_j} = \nu_j + \sum_a (I_{j,a} - \mathbb{E} I_{j,a})(\delta_a - \delta_\partial)$  gives, after tensor expansion, centered moment coefficients on subsets  $S$ . The signed tensor has zero-mass supremum norm  $2^{|S|-1}$ . Singletons vanish. Expand each centered moment into cumulant partitions, with blocks of size at least two. If  $b_B = 2^{|B|-1} \sum_a |\text{cum}(I_{j,a_j} : j \in B)|$ , a partition into  $m$  blocks contributes at most  $2^{m-1} \prod_B b_B$ . Sum disjoint block collections, then drop disjointness to obtain  $\frac{1}{2} \sum_{m \geq 1} (2 \sum_B b_B)^m / m! = (e^{2\mathcal{K}} - 1)/2$ . The identical finite partition expansion with weights  $t^{|B|}$  proves (G.24). Multilinearity gives  $2\mathcal{K} \geq \sum_{|S| \geq 2} 2^{|S|} |\text{cum}(B_j : j \in S)|$ . When  $2 \max_j r_j < 1$ , all factors in  $F_2$  are positive, so taking logarithms proves the last assertion. No analytic infinite cumulant series is assumed.  $\square$

The law on the right of (G.23) is the product of categorical marginals, not yet independent Poisson coordinates. Poissonizing occupied mass  $s_j \leq p$  costs at most  $\sum_j s_j^2 \leq np^2$  by coupling the total count and using the same type kernel. The finite inequality remains true, but its global-smallness hypothesis fails as follows.

**Theorem G.9** (Small microscopic distance with divergent absolute activity). *Let  $q$  tend to infinity through odd primes and set*

$$a_q = \lfloor \log_2 q \rfloor, \quad M_q = 2^{q-1+a_q}, \quad L_q = q - 1.$$

Take  $\mathcal{A}_{M_q}$  to be the whole space. If the retained sets have the exact low-type marginals and omit  $o(M_q/\log M_q)$  sites, then

$$\liminf_{q \rightarrow \infty} \frac{\log M_q}{M_q} \mathcal{K}_{\text{abs}, M_q} \geq \frac{\log 2}{2} (\log 3 - 1) > 0. \quad (\text{G.25})$$

This applies to  $G_0$  of Appendix F and to the stronger retained sets in Appendix G. The two-site contribution tends to zero, so the activity of orders at least three diverges. Nevertheless  $\Delta_{M_q}^{\text{mic}} \rightarrow 0$ , already within the earlier sufficient labelled range.

*Proof of Theorem G.9.* For  $M = M_q$ ,  $L = q - 1$ ,  $a_q = \lfloor \log_2 q \rfloor$ ,

$$H \sim \beta q, \quad \Lambda = 2^{a_q} - (q - 1)2^{-(q-1)} \asymp q.$$

Thus  $a_q = o(\log M)$  and both  $\log \Lambda$  and  $\log \Lambda + \log(1 + \Lambda)$  are  $O(\log H) = o(V - c\nu)$  for every fixed  $c$ . Let  $g = |G|$ ,  $S_* = 1 - 2^{-E_*-1}$  and  $B_j = \sum_{\alpha \in \mathcal{T}_*} I_{j,\alpha}$ . Their common exact mean is  $r = pS_* \leq p$ , and  $gr \leq \Lambda$ .

Consider the event

$$\mathcal{E}_q = \{f(q) = -1, f(\ell) = 1 \text{ for every prime } \ell \leq M + Q, \ell \neq q\}.$$

Its probability is  $2^{-\pi(M+Q)}$  in the original unconditioned model; it is not a new admissible conditioning event. The cutoff  $M + Q$  covers the queried vertices and has  $Q = o(M)$ . A cylinder enlarged to  $2M$  would change the witness cost and cannot be substituted here. On this event every queried value is  $(-1)^{v_q(m)}$ . For  $(t + 1)q \leq M$  and  $q \nmid t(t + 1)$ , the endpoints  $tq, (t + 1)q$  are negative and the  $q - 1$  interior values are positive. Therefore the site  $j = tq$  carries the low type  $(0, +1)$ . With  $T = \lfloor M/q \rfloor - 1$ , the number of certified indices is exactly

$$T - \lfloor T/q \rfloor - \lfloor (T + 1)/q \rfloor = (1 - 2/q)M/q + O(1).$$

The omissions are counted in  $[1, M]$ , including the  $L$  terminal indices outside  $G \subseteq \{1, \dots, M - L\}$ . Their additional cost  $L = O(\log M)$  is negligible. Omitting  $o(M/H) = o(M/q)$  sites leaves at least  $N_0 = (1 - o(1))M/q$  retained low-type atoms. For  $N = \sum_{j \in G} B_j$  and sufficiently large  $q$ ,

$$F_2 = (1 - 2r)^g \mathbb{E} \left( \frac{3 - 2r}{1 - 2r} \right)^N \geq 2^{-\pi(M+Q)} (1 - 2r)^g \left( \frac{3 - 2r}{1 - 2r} \right)^{N_0}.$$

The prime number theorem gives  $\pi(M + Q) = (1 + o(1))M/H$ ;  $g \log(1 - 2r) \geq -O(\Lambda)$ ; and  $\log((3 - 2r)/(1 - 2r)) = \log 3 + O(r)$ . Since  $\Lambda \asymp H = o(M/H)$ , it follows that

$$\log F_2 \geq \{\beta(\log 3 - 1) + o(1)\}M/H.$$

Lemma G.8 proves (G.25). The omission condition holds for  $G_\vartheta$  by (G.6) and for  $G_0$  by its smaller deletion. Proposition G.7 gives the vanishing pair layer, and hence divergence of the remaining orders. These parameters lie even in the graph-based labelled budget. In particular [Theorem 7.7](#) gives the asserted vanishing microscopic distance, independently of the activity calculation.  $\square$

For a fixed  $t > 0$ , the same witness has leading lower exponent  $\beta\{\log(1 + t) - 1\}M/H$ . It diverges when  $t > e - 1$ ; this is a threshold for the witness, not an optimal boundary of the model. Restricting the occupancy to  $N \leq b\Lambda$ ,  $b > 1$ , can have vanishing ordinary total-variation cost, but it changes its means, cumulants, and conditional forcing identities. No such nonlocal truncation is used to retain an invalid absolute-smallness claim.

## Availability and declaration

The scope, explicit literature inputs and limitations of the Lean development, together with its seven Palomar registrations, are described in the article’s section “[Formalization and Palomar registrations](#)”. The article also records the [use of AI assistance and the author’s responsibility](#). This companion and the article form one proof package, with their sources and bibliography available together.

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